

Leaky Sanctions^{*}

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Abstract

The sanctions imposed on Russia in 2022 are the most extensive in modern history, yet at least 26% of pre-war EU exports of sanctioned products continued to reach Russia through third-country intermediaries. Standard quantitative trade frameworks, however, assume full enforcement and cannot quantify the consequences of evasion. Combining Russian transaction-level customs records with a new granular dataset on EU export bans, we document the mechanisms and magnitude of sanctions evasion and develop a quantitative trade model with endogenous transshipment to assess its consequences. Our analysis yields three main insights. First, we estimate that the export bans increased effective trade costs from the EU to Russia by about 27% on average, well below the prohibitive levels implied by full enforcement. This observed level of evasion reduces Russia's real income loss by 23% relative to the full-enforcement scenario. Second, pre-war trade costs predict which countries become transshipment hubs but understate the true cost of evasion: when evasion costs consist only of the additional trade cost from rerouting, Russia's real income loss falls to just 34% of the full-enforcement level. This implies that the sanctioning coalition created substantial additional barriers beyond the costs of rerouting itself. Third, closing just five transshipment hubs in the model reduces imports by an additional 8 pp and increases Russia's real-income losses by 25%.

Keywords: Evasion, Sanctions, Export Bans, Transshipment, Geoeconomics.

JEL-Classification: F11, F13, F14, F51, F52.

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1 Introduction

Russia’s invasion of Ukraine in February 2022 triggered the most comprehensive sanctions episode in modern history. A broad coalition consisting of the European Union (EU), the United States, and other allies froze assets, severed financial ties, and banned thousands of products from export to Russia, from advanced semiconductors to champagne and luxury cars. The immediate objective of the export bans was clear: to cut off Russian access to Western goods. But whether such bans are successful depends on how easily they can be evaded. This paper measures the extent of sanctions evasion by exploiting transaction-level customs data and granular information on export bans, and quantifies its consequences for the effectiveness of the sanctions regime in a trade model.

Measuring evasion is challenging because the underlying activities are illicit and leave little trace in standard trade data. However, in our setting, we can directly observe one important form of evasion: transshipment through third countries. Russian transaction-level customs data report both the country of origin and the country of dispatch for each import shipment, allowing us to identify coalition-origin goods that enter Russia via third countries after the bans take effect. We focus on the EU’s export bans, for which we compile a novel product-level dataset that identifies the precise scope and timing of each restriction.

The EU export bans prohibit both direct and indirect exports to Russia and are enforced by national authorities in each EU member state through substantial penalties, including criminal liability for executives of noncompliant firms. Consistent with this legal framework, direct EU exports of sanctioned products to Russia fall to zero in official trade statistics after the bans take effect. Because indirect exports are also prohibited, EU firms have little incentive to misreport the country of origin when exporting to third countries, as doing so would expose them to liability. We therefore define evasion as the continued import into Russia of EU-origin products after the export bans are in force. Matching our new sanctions dataset to the Russian customs records, we observe evasion directly and find that at least 26% of the EU’s pre-war monthly export value of products fully covered by the bans continues to enter Russia after their implementation. While evasion may occur through additional channels, including misclassification or false origin declaration, these are not observed in the data. Our measure for evasion should therefore be interpreted as a lower bound.¹

Evasion occurs through two distinct transshipment modes. Under re-exporting, intermediaries in third countries purchase sanctioned goods from EU exporters and ship them onward to Russia, becoming the seller of record while the product’s origin remains un-

¹ Transshipment through third countries has been documented in other trade policy contexts; see, for example, *Financial Times*, “Chinese exporters ‘wash’ products in third countries to avoid tariffs,” May 4, 2025, <https://www.ft.com/content/147fddbb-7031-4347-9251-4425614e138d>

changed. For example, Ferrari cars, produced exclusively in Italy, were exported directly to Russia before the war but subsequently reached Russia via Belarus or the United Arab Emirates, with Italian origin preserved but the exporting firm shifting from Italian manufacturers to intermediary companies. Under in-transit transshipment, exporters nominally declare a third country as the final destination, such as Kazakhstan or Belarus, while the goods enter Russia and never continue onward. Both modes exploit gaps in the enforcement of export controls: reexporting circumvents the ban by changing who sells, while in-transit transshipment circumvents it by misrepresenting the ultimate destination of goods. In-transit transshipment dominates immediately after the bans enter into force in July 2022, reflecting the ease with which destination declarations can be falsified at short notice. As enforcement tightens, with stricter rules curbing transit flows beginning in mid-2023, reexporting rapidly displaces in-transit transshipment as the primary evasion mode. This dynamic suggests that sanctions evasion is not static: intermediary networks adapt as enforcement closes off particular channels.

Re-exporting is highly concentrated in a small number of hubs. The majority of re-exported EU-origin, export-banned goods pass through just three countries—Türkiye, Belarus, and China—with Türkiye alone accounting for more than half of the total re-exported value. At the same time, re-exporting is not confined to these hubs: nearly one-fifth of re-exported trade is routed through a broader set of intermediary countries. Within this group, activity is itself concentrated, with the United Arab Emirates, Uzbekistan, and Serbia emerging as the most important secondary hubs.

Evasion differs substantially across products. Among the 688 sanctioned HS6 products subject to a full export ban by July 2022, the ratio of post-sanctions to 2019 average monthly imports varies widely, and 15% of products are imported at higher values after the bans than before. To understand this heterogeneity, we relate evasion shares to pre-war product characteristics. Evasion is systematically higher for goods whose shipments can be more easily concealed within normal trade flows. This includes goods with many alternative foreign suppliers outside the EU, where existing export relationships either provide cover through larger volumes of legitimate trade or reflect established know-how in handling these products. Evasion is also higher for goods that already moved, even minimally, through intermediaries before the bans, which eliminates the costs of establishing new routes. In addition, goods that are not bulky and therefore cheaper to divert, and goods purchased by a wide set of Russian buyers, exhibit higher evasion, consistent with lower detection risk and greater ease of concealment.

The products that experience the largest increases in post-sanctions imports (i.e., those exceeding their 2019 levels) display a different pattern. These goods tend to have relatively few alternative suppliers and little prior third-country routing, suggesting that the difficulty

of substitution dominates: for these products, Russia’s need to secure continued access outweighs the cost of evasion.

To quantify the consequences of export bans and evasion for trade patterns and real income, we develop a general-equilibrium model of international trade with endogenous transshipment. A key feature of the framework is that it distinguishes between legal trade diversion, i.e., substitution toward non-sanctioning suppliers, and evasion through intermediaries. This separation is essential because the two channels have very different implications for both enforcement and welfare: diversion reflects successful trade reallocation, whereas evasion circumvents the bans. The model allows us to quantify the trade and income effects of the export bans and their evasion while accounting for general equilibrium adjustments, to study the mechanisms behind evasion, and to compute different counterfactual scenarios, such as full enforcement, closing specific transshipment hubs, or enlarging the coalition size.

The environment builds on the multi-product version of the [Eaton and Kortum \(2002\)](#) model (henceforth, EK). There are multiple products in the model, each composed of a continuum of tradable intermediate varieties produced by perfectly competitive firms. To measure transshipment opportunities, we rely on a well-known no-arbitrage condition: the triangle inequality (TI) of trade costs. The TI states that if the export cost we assign to the shipment from country e to importer i exceeds the cost of shipping a product from country e to i via a third country h , country h would become an intermediary hub (see [Foellmi et al., 2024](#)). In turn, this implies that when we impose a ban on direct exports from e to i , the model suggests which third country h becomes the intermediary hub.

We calibrate the model so that each product corresponds to an HS6 product in the 2019 UN Comtrade trade data. We develop a new calibration strategy to recover domestic product-level expenditure, which is typically not available at such disaggregated levels. Quantifying the model at the product level offers several advantages relative to the more aggregated approaches that are standard in the literature. First, it allows us to impose export bans directly at the level at which they are defined, whereas sector-level quantification would require ad hoc aggregation because not all products within a sector are restricted. Second, the model generates transshipment patterns at the origin-product level, which we can directly compare to our disaggregated customs data. Third, this introduces product-specific trade dependencies and elasticities to better approximate the substitutability of an origin-product cell.

The EK framework delivers a gravity equation in which bilateral exports depend on trade costs, productivity, and multilateral resistance terms. To recover the de facto increase in trade costs induced by the export bans, we estimate the model-implied gravity equation using a triple-difference specification at the HS6–exporter–month level over 2019 and 2022–2024. The specification compares (i) products subject to the export bans with

unaffected products, (ii) periods before and after the bans, and (iii) EU exports to Russia with exports to Russia from non-sanctioning countries.

Our specification includes finely grained fixed effects with a natural structural interpretation. Product–exporter fixed effects absorb time-invariant comparative advantage and geography; exporter–month fixed effects capture, among other factors, origin-wide changes such as reputational risk or financial and logistical restrictions (including SWIFT exclusion); and product–month fixed effects correspond to time-varying multilateral resistance on the importer side and controls for shifts in Russian demand at the product level. As is standard in the trade literature, we estimate the specification using PPML, which is well-suited to the multiplicative structure of gravity and, crucially for our setting, can handle the extensive margin, where much of the adjustment to the export bans occurs (Head and Mayer, 2014; Santos Silva and Tenreyro, 2006; Yotov et al., 2016). Our baseline estimate implies that, following the export bans, EU exports of products covered by the bans to Russia fell by roughly 80%, relative to products not affected by the export bans and to exports from non-coalition countries. Through the lens of our gravity model, and using trade elasticities from Fontangé et al. (2022), this implies that effective bilateral trade costs on EU exports of banned products to Russia increased by about 27% on average—a large increase, but far short of the prohibitive level implied by perfect enforcement.

Our counterfactual analysis shows that incomplete enforcement substantially attenuates the trade and welfare effects of export bans. We compute several counterfactual scenarios using the calibrated model, varying the degree of enforcement among coalition countries. As a benchmark, we quantify the consequences of fully enforced export bans, which correspond to the standard quantitative exercise in the literature. Under full enforcement, Russia’s real income declines by 1.20% relative to a world without the bans, and exports of sanctioned products by coalition countries fall to zero. We then consider a counterfactual scenario with incomplete enforcement. Using our DiD estimates, which capture the implied evasion behavior in the data, we impose a 27% increase in effective trade costs induced by the coalition’s export bans. The quantification results in a 71% reduction in Russia’s imports of sanctioned products from coalition origins and a 23% smaller real income loss relative to the full-enforcement scenario. From a policy perspective, fully enforcing the export bans would have increased Russia’s income losses by about 30%.

Allowing exporters to endogenously reroute trade through intermediary countries shows that pure transshipment costs alone cannot fully account for the observed import decline. In a third counterfactual, we allow for endogenous transshipment once direct coalition–Russia export links are shut down, with trade rerouted through intermediary (non-coalition) countries along the cheapest indirect links. In this scenario, the only increase in trade costs for EU exports to Russia arises from additional transportation costs associated with

re-exporting. The model implies that Russia’s imports of banned products drop by 39%, and its real income loss is 66% lower than under full enforcement. This indicates that while pure transshipment costs are substantial, they are not sufficient to account for the import decline observed empirically. Finally, we consider a counterfactual in which key transshipment hubs are closed: blocking re-exports from the five main hubs in the model reduces imports by an additional 8 pp and increases Russia’s real-income losses by 25% relative to the full transshipment scenario.

The framework’s predictions for transshipment hubs align closely with observed patterns in the data. This constitutes a particularly demanding out-of-sample test, as the model is calibrated using only pre-war trade data from 2019 and does not rely on the Russian customs data. In the data, the five main transshipment hubs are Belarus, China, Serbia, Türkiye, and the United Arab Emirates, while in the model UAE is replaced by India (which is the sixth most important hub in the data). The correlation between how often a given hub is used for a given origin–product pair in the model and in the data is 0.83 when excluding Türkiye (an outlier in the data) and 0.70 when Türkiye is included. Transshipment is highly concentrated and occurs primarily through this small set of hubs in both the data and the model.

We then examine why actual hub use differs from what the model predicts. Geopolitical alignment with the sanctioning coalition is largely unrelated to this gap. Instead, the evidence points to economies of scope: in the data, the intensity with which EU origins use a specific hub is nearly identical for banned and non-sanctioned products, with a correlation of 0.99. This implies that legal export flows are highly predictive of how illicit trade is rerouted.

We contribute to several strands of the literature. First, we contribute to work on the circumvention of trade barriers. A large literature studies evasion in the context of taxes (Fisman and Wei, 2004; Laffitte and Toubal, 2022) and tariffs (Fisman et al., 2008; Javorcik and Narciso, 2008), with renewed interest following the U.S.-China trade war (Che et al., 2025; Freund, 2025; Iyoha et al., 2024) and the recent sanctions against Russia (Chupilkin et al., 2025, 2023b; Scheckenhofer et al., 2025; Tyazhelnikov and Romalis, 2024). We add to this literature in two distinct ways. In contrast to existing work, we can directly measure evasion in customs trade data rather than infer it from trade reallocation patterns. Moreover, we quantify the consequences of evasion for the effectiveness of export bans in a quantitative general-equilibrium framework, whereas most of the existing evidence in this strand is reduced-form and silent on welfare.

Second, we contribute to the growing literature on the trade and income effects of economic sanctions, particularly those imposed on Russia. Existing work studies these effects using firm-level data (Corsetti et al., 2025; Crozet and Hinz, 2020; Crozet et al., 2021; Egorov et al., 2025), more aggregated trade data (Egorov et al., 2025), and quantitative

general-equilibrium trade models (Chowdhry et al., 2024; Flach et al., 2024; Hausmann et al., 2024; Hinz and Monastyrenko, 2022). Rather than focusing solely on the effects of sanctions on trade flows or welfare under full enforcement, we study evasion and, through it, the effectiveness of sanctions. Combining an event-study empirical design with a quantitative general-equilibrium framework, we measure evasion directly and quantify how it attenuates the trade and income effects of export bans. Existing general-equilibrium quantifications abstract from evasion entirely, while reduced-form studies that discuss evasion lack the detailed data needed to distinguish between partially and fully banned products, a distinction our new dataset provides and that is essential for measuring evasion precisely.

Third, we contribute to the measurement of trade sanctions. We compile a uniquely detailed, tariff line-level dataset on EU export bans that identifies which products are affected, the precise timing of each restriction, and whether bans apply fully or only to specific technical characteristics. Constructing such data is non-trivial because the EU’s sanctions regime varies widely across products and over time, often within narrowly defined tariff lines. For example, within a single tariff line covering industrial pumps, only certain models—such as those suitable for chemical-weapons production—are restricted, while others remain exportable. As a result, more than half of tariff lines are initially only partially restricted rather than fully banned.² Existing sanctions datasets either code sanctions at the country-pair level, with the goal of covering all sanctions worldwide and across all sanction types, not only trade sanctions (Felbermayr et al., 2024; Morgan et al., 2023), or include product-level information without resolving the precise legal scope of restrictions within tariff lines, as in commonly used sources such as the Global Trade Alert or EU Council regulations (Crozet and Hinz, 2020; ?). As a consequence, they would misclassify a large volume of legal shipments as illicit and mechanically overstate evasion. We overcome this limitation by identifying, for each CN8 product, whether restrictions apply fully or only to specific technical specifications, as well as the exact date at which they become binding, based on TARIC and a comprehensive review of the European Commission’s legal gazettes.

Finally, we contribute to the development of quantitative trade models (see Caliendo and Parro, 2022; Costinot and Rodríguez-Clare, 2014, for an overview). Specifically, we are the first to quantify an Eaton and Kortum (2002)-type framework at the 6-digit level (while allowing for endogenous transshipment), using only readily available data, with product-specific trade costs and trade elasticities.

The paper proceeds as follows. Section 2 introduces the data construction and describes how we leverage our new data to measure evasion. Section 3 presents descriptive facts

² Partial sanctions reflect the EU’s “smart sanctions” approach, which aims to restrict only items with sensitive or military applications while minimizing collateral damage to legitimate trade.

on the magnitude and patterns of evasion. Section 4 develops a model to interpret these findings. Section 5 outlines the triple DiD empirical strategy and reports the main results, and Section 6 provides counterfactual analyses and quantifies the welfare consequences. Section 7 concludes.

2 Measuring Evasion

The full-scale Russian invasion of Ukraine in February 2022 marked the largest military conflict in Europe since World War II. In response, Western allies, led by the U.S. and the EU, imposed an extensive and evolving sanctions regime against Russia, spanning multiple policy domains including finance (e.g., exclusion from SWIFT and asset freezes), trade (import and export bans), transport (airspace restrictions), energy (oil and gas), and targeted sanctions on firms and individuals. In this paper, we focus on EU export bans, which form a central pillar of the sanctions regime and are particularly well suited for studying evasion.³ The scope of the export bans expanded substantially over time as successive sanctions packages were adopted, ultimately covering a broad range of products (Table A1).

Because the bans apply to both direct and indirect exports, EU firms face strong incentives to comply formally when exporting banned products to third countries: misreporting origin or misclassifying products would constitute a direct sanctions violation. Russia does not enforce EU export bans, so Russian importers face no enforcement-related incentives to misreport the country of origin in customs declarations. Customs authorities therefore continue to record the declared country of origin for imports, including shipments that enter Russia via third countries. As a result, goods that enter Russia despite the export bans often retain their EU origin in customs records, making transshipment and hence evasion directly observable in the data.⁴ Our analysis focuses on this observable margin of evasion. To the extent that evasion occurs through relabeling of origin or product codes by intermediary firms, it is not directly observed in the data, and our estimates should therefore be interpreted as a lower bound. This feature distinguishes export bans enforced at origin from import tariffs or import bans, where evasion operates through misreporting at the border and is unobservable in customs data.

³ We refer to both outright export prohibitions and export licensing requirements as *export bans*. Although the EU’s sanctions regime distinguishes between the two, licenses for restricted exports to Russia are rarely granted outside narrowly defined exemptions (e.g., humanitarian uses). In practice, licensing requirements therefore function as bans.

⁴ Under Russia’s parallel-importing regime, imports of branded goods are permitted without the authorization of the trademark holder, including goods originating in sanctioning countries. Russian customs authorities therefore continue to accept and record the original country of origin for such shipments.

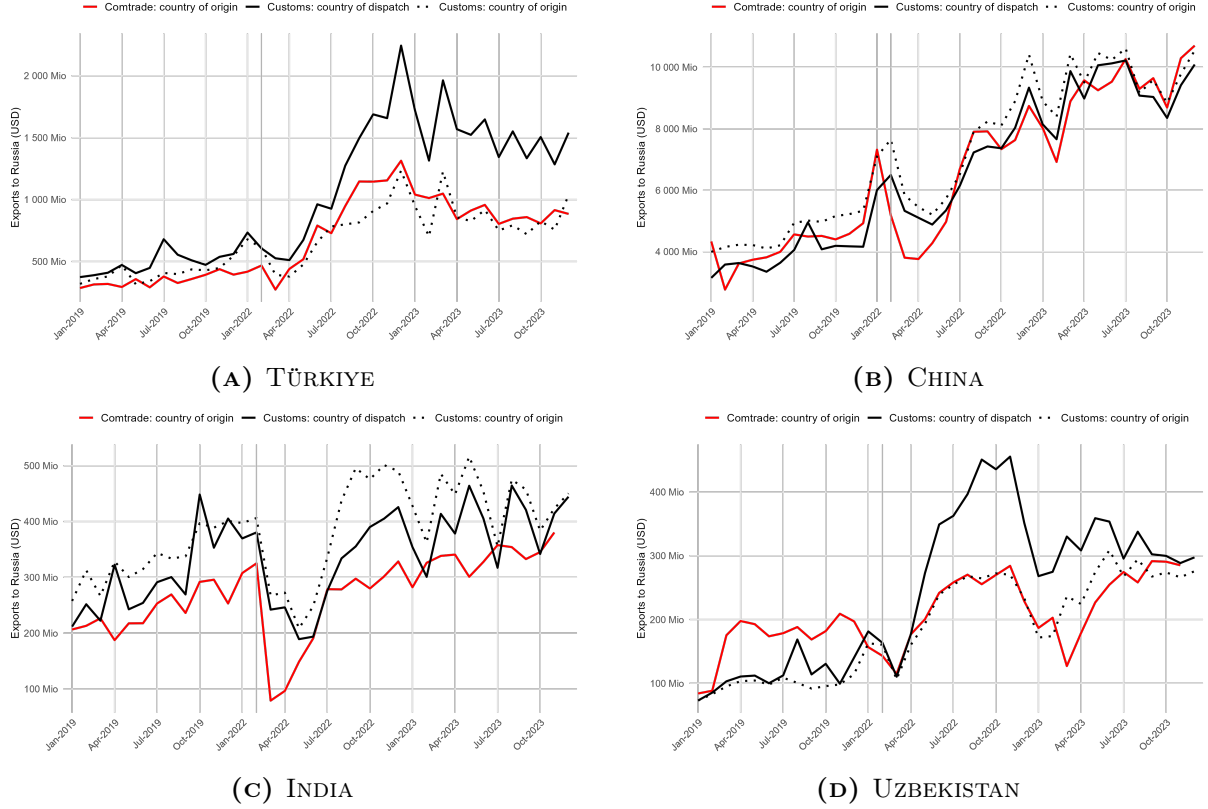
2.1 Russian Customs Trade Data

To observe evasion through transshipment that does not involve manipulation of the true origin, we use confidential Russian transaction-level customs data covering the universe of import declarations. These data report firm-to-firm transactions and include detailed information on product descriptions, 10-digit product codes, quantities, values, and, crucially, the declared country of origin for each shipment. Furthermore, the data distinguish between the country of origin—where the product was produced—and the country of dispatch—the location from which the product was last shipped to Russia. Prior to the war, these customs records formed the basis for Russia’s official aggregate trade statistics reported through UN Comtrade and domestic statistical releases. Other work using these data show that they align closely with other trade statistics (Babina et al., 2026; Chupilkin et al., 2023a; Egorov et al., 2025; Korovkin and Makarin, 2023).

A natural concern is whether Russian customs data remain reliable in the post-invasion period. Systematic manipulation of customs microdata is considerably more difficult than manipulation of aggregate statistics, such as GDP or headline trade balances, because of the high dimensionality of the data. In 2023 alone, the customs records contain roughly 30 million individual transactions. Altering these data in a consistent manner across products, firms, origins, and time would be substantially more complex than adjusting a small number of aggregate indicators.

We further assess data quality by comparing Russian import records to mirror export data from UN Comtrade, which report exports to Russia by country of origin as declared by partner countries. Such comparisons are uninformative for sanctioning countries, for which discrepancies are expected by construction and are precisely how we measure evasion. We therefore focus on non-sanctioning countries, whose exports to Russia remain legal and unrestricted and for which there is no incentive to misreport trade. If Russian customs data were systematically manipulated, imports from these countries would also diverge from partner-reported exports. We examine the four largest post-invasion exporters to Russia outside the sanctioning coalition for which UN Comtrade data are available; additional examples are reported in Appendix Figure A2.

Figure 1 shows that, for Türkiye, China, and Uzbekistan, UN Comtrade mirror exports closely track Russian customs imports by country of origin. For India, Comtrade flows are systematically lower, but this pattern predates the full-scale invasion of Ukraine in February 2022 and likely reflects known limitations of mirror statistics, such as valuation differences between CIF and FOB prices or reporting thresholds, rather than manipulation of the customs data. Overall, these comparisons suggest that the Russian customs data reliably capture trade flows and origin information.



Note: The figure compares Russian customs imports with mirror exports to Russia reported in UN Comtrade, by month, year, and partner country. Data are shown for 2019 and for 2022–2023 for the largest exporters to Russia outside the sanctioning coalition, for which UN Comtrade data are available. Trade values are expressed in millions of U.S. dollars.

FIGURE 1. VALIDATION OF RUSSIAN CUSTOMS DATA

2.2 Data on EU Export Bans

Information on origin alone is not sufficient to measure evasion; doing so also requires knowing which EU products are legally subject to export bans and when these restrictions take effect. We therefore compile a detailed dataset on EU export bans that identifies the affected products and the timing of the restrictions. Constructing such a dataset is non-trivial because export bans are often defined at a level of technical detail finer than the CN8 classification used in EU trade statistics and therefore apply only to specific product variants within a tariff line. For example, within a given CN8 category of industrial pumps, only variants with technical characteristics suitable for chemical weapons production may be restricted, while otherwise similar pumps remain legally exportable. This reflects the logic of “smart sanctions”, which aim to restrict access to strategically sensitive inputs while limiting collateral losses for EU exporters. If legally exportable product variants are mistakenly classified as banned, any continued exports of these products would be incorrectly counted as evasion. Accurate identification of the legal scope of export bans is therefore essential for our analysis.

We combine information from EU Council regulations with the EU’s Integrated Tariff (TARIC) to recover all trade restrictions affecting EU–Russia trade and their type. TARIC records all EU-measures governing the import and export of goods, including tariffs, documentation requirements, and import and export restrictions.⁵ The database is updated daily and covers all CN8 product codes—over 10,000 products—since 1994. Using TARIC, we retrieve all export restrictions affecting trade with Russia at a daily frequency. When sanctions are defined at a more aggregated level, for example, the February 2022 ban on airplane exports under the two-digit product code 88, all CN8 codes within that category inherit the restriction.

TARIC does not distinguish between partial and full export bans; rather, it indicates whether a given product is subject to one or more export restrictions at a given point in time. We distinguish between fully and partially banned products by combining TARIC with a manual review of the relevant EU Council regulations and their annexes. Products are classified as partially banned for two reasons. First, EU Council regulations may specify *ex* CN codes, which restrict only specific product variants within a CN8 category. Second, products covered exclusively by the EU’s dual-use regulation are restricted only for variants meeting specific technical characteristics and are therefore classified as partially banned. Importantly, some products covered by the dual-use regulation are also subject to additional export restrictions. In these cases, all variants within the CN8 category are restricted, and the product is classified as fully banned. TARIC allows us to identify such overlap across legal bases and to distinguish within the set of dual-use products those that are partially versus fully banned.

This distinction is most relevant early in the sanctions regime, when relatively few products were fully banned. Following the second sanctions package of February 25, 2022, export restrictions applied primarily to dual-use goods, aviation products, and oil-refining technologies, leaving many products only partially covered. As subsequent sanctions packages were adopted, the scope of full export bans expanded, and several products initially classified as partially banned became fully banned as entire CN8 categories were explicitly banned. We track these changes over time to maintain a consistent product-level classification.

Furthermore, the correct entry-into-force date can only be retrieved through TARIC. A common approach in the literature, relying on the published EU Council regulations and their annexes, uses the regulation’s publication date as the entry-into-force date for export bans (e.g., [Crozet and Hinz, 2020](#); ?). However, this is often inaccurate, as entry-into-force dates do not necessarily coincide with the publication date of the regulation. For example, a regulation published on February 25 may ban the export of oil-refining technology, yet

⁵ https://taxation-customs.ec.europa.eu/customs-4/calculation-customs-duties/customs-tariff/eu-customs-tariff-taric_en

continue to allow exports under contracts concluded before that date until May 27. Such phased implementation and exemptions are difficult to capture when relying only on the legal texts. In contrast, TARIC records unambiguous, product-specific entry-into-force dates.

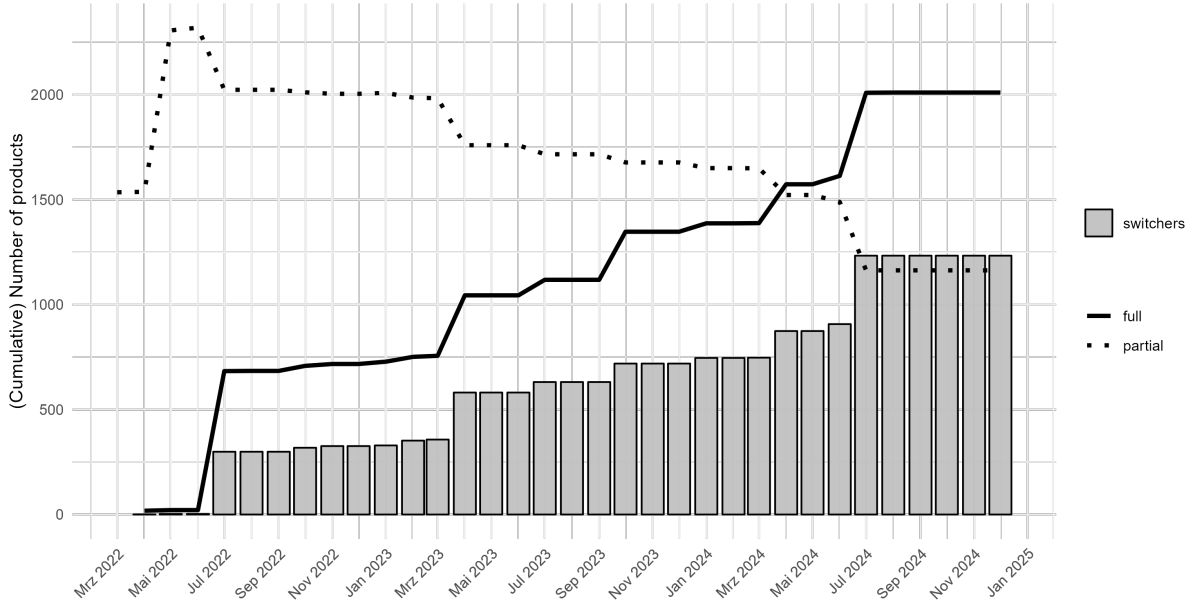


FIGURE 2. EU EXPORT BANS OVER TIME

Note: The figure shows the cumulative number of HS6 products covered by EU export bans over time. The solid line denotes fully banned products, the dotted line partially banned products (accounting for transitions from partial to full coverage), and the gray bars indicate the number of products that switch from partial.

To combine the sanctions data with the Russian customs data, we aggregate products to the six-digit Harmonized System (HS6) level. An HS6 product is classified as fully sanctioned if all underlying CN8 products are sanctioned, and as partially banned otherwise. We express all products in the 2017 revision of the HS nomenclature (HS2017).⁶ At the monthly frequency, a product is coded as affected by the export bans only if the restriction is in force for at least 15 days of the month.

By July 2022, 51% of all 6-digit products were subject to EU export bans, driven predominantly by partially sanctioned products (partial: 2,023 products; full: 688 products). By the end of 2024, this pattern had reversed: 38% of all products were fully sanctioned, while only 23% remained partially sanctioned (partial: 1,163 products; full: 2,010 products). Figure 2 illustrates that this shift reflects both the expansion of the sanction regime and the progressive tightening of existing sanctions. In particular, the EU increasingly

⁶ We use standard concordance tables provided by the United Nations: <https://unstats.un.org/unsd/classifications/Econ>. HS2017 products can map to multiple HS2022 codes, in which case the HS2017 product is classified as partially sanctioned if some underlying HS2022 codes are not fully banned. This coding choice avoids mechanically overstating the scope of export bans.

targeted the remaining unsanctioned CN8 codes within affected 6-digit product categories, effectively converting roughly 1,200 products from partial to full sanction status over the period of almost three years.

Figure A3 shows that the transition from partial to full export bans is concentrated in a subset of sectors. The shift is particularly pronounced in sectors where restrictions were initially defined through textual provisions, most notably dual-use and high-tech goods in Optics and in Arms & Ammunition. A similar evolution is observed in sectors where sanctions first applied narrowly to specific CN8 products rather than to entire two- or four-digit categories, such as Machinery & Electrical Equipment, and Plastics. By contrast, sectors such as Jewelry and Works of Art exhibit little change in the share of partially sanctioned products over time. In these categories, product-level conditional exemptions for luxury goods persisted, limiting the scope of full export bans.

EU export bans have a sharp and immediate effect on direct trade. As shown in Figure 3, once products become fully subject to export bans, direct EU exports to Russia reported in official EU trade statistics (Comext) drop to zero. By contrast, exports of partially banned products also decline but do not collapse, consistent with these goods remaining largely legally exportable. Exports of non-banned products also decline following the invasion, indicating that the contraction in EU–Russia trade is not driven by export bans alone: Broader sanctions, heightened uncertainty, the retrenchment of multinational firms driven largely by reputational concerns, and weakening Russian demand all contribute to a generalized reduction in direct trade.

2.3 Identifying Evasion in Customs Data

We define evasion as the continued import into Russia of products that are fully subject to EU export bans after the bans take effect. A transaction is classified as evasion if it involves a fully banned product, lists an EU member state as the country of origin in Russian customs data, and occurs in a month in which direct EU exports of that product to Russia are prohibited. Because direct EU exports of fully banned products drop to zero after the bans enter into force, any such imports must occur indirectly. Unlike tariffs or import bans, this setting allows evasion to be observed in customs data.

An example illustrates the definition. In 2019, Russia imported 44 Ferrari cars, all recorded with Italy as the country of origin. In 2023, after exports of such cars to Russia were prohibited, six Ferrari cars still entered Russia, again with Italy listed as the country of origin.⁷ Under our definition, these transactions are classified as evasion.

Russian customs data separately report the country of origin and the country of dispatch. This allows us to identify transshipment routes and to distinguish between different forms

⁷ Product descriptions indicate that these were new Ferrari cars rather than used vehicles.

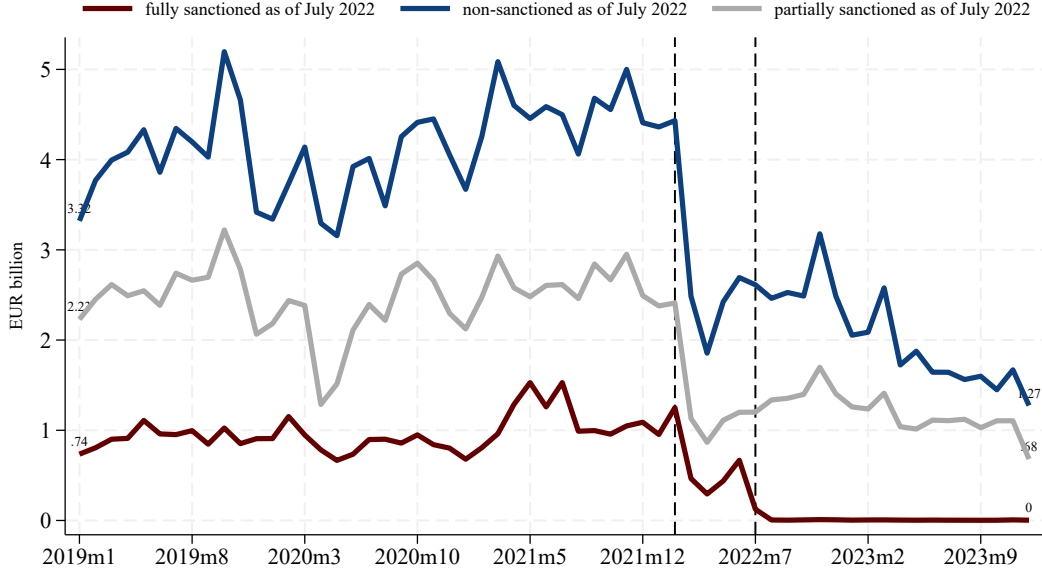


FIGURE 3. DIRECT EU EXPORTS TO RUSSIA

Note: Panel (A) shows monthly EU exports to Russia by export-ban status using official EU trade statistics (Comext), with values in billion euros. Products are classified based on their legal status as of July 2022 into fully banned, partially banned, and non-banned categories.

of indirect trade. In the Ferrari example, the country of dispatch shifts from Italy before the bans to third countries after the bans, while the reported country of origin remains Italy. More generally, we use the country of dispatch to trace intermediary countries.

3 Stylized Facts about Evasion

Using our newly assembled data, we document five stylized facts about the magnitude and patterns of evasion of EU export bans on Russia. Together, these facts characterize both the extent of evasion and the channels through which it occurs, and they guide the structure of the quantitative model developed in the next section.

Fact 1: Banned EU exports continue to reach Russia Figure 4 plots monthly Russian imports of EU-origin products that subsequently become subject to EU export bans, measured in values and quantities. Import values rise in the months between Russia’s invasion of Ukraine in late February 2022 and the introduction of the export bans in July 2022, before falling sharply once the bans enter into force. This increase prior to July 2022 is consistent with stockpiling in anticipation of tighter restrictions, which is plausible given that the first sanctions were announced shortly after the invasion. The largest adjustment occurs with the introduction of the export bans in July 2022, when imports drop sharply.

Despite this decline, Russian imports of EU-origin goods do not fall to zero, as would be implied by fully enforced export bans. Instead, they stabilize at a positive level and remain broadly flat through the end of 2024. The persistence of imports is not driven solely by price adjustments: quantities display a similar pattern. Since official EU export statistics report zero exports of these products after the bans enter into force, the continued appearance of EU-origin goods in Russian customs records necessarily reflects transactions that circumvent the export bans (Figure 3). As a simple summary measure of evasion, we compute the ratio of average monthly imports after July 2022 to average monthly imports in 2019. This ratio equals 26.0% for import values and 12.4% for quantities. These estimates constitute a lower bound on evasion: they exclude partially banned products, for which evasion may be easier because very similar non-banned variants exist, and they capture only transactions in which the EU origin remains visible in the data, thus not accounting for any origin relabeling.

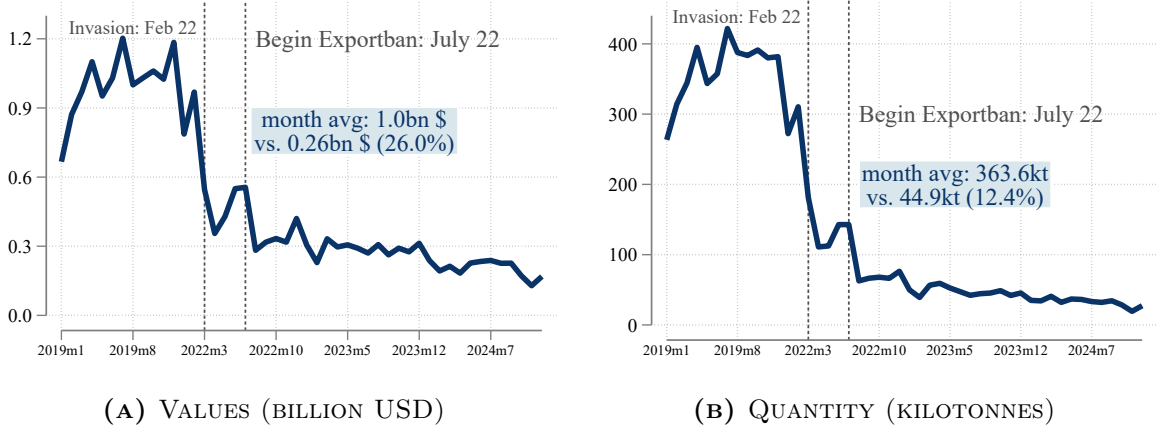


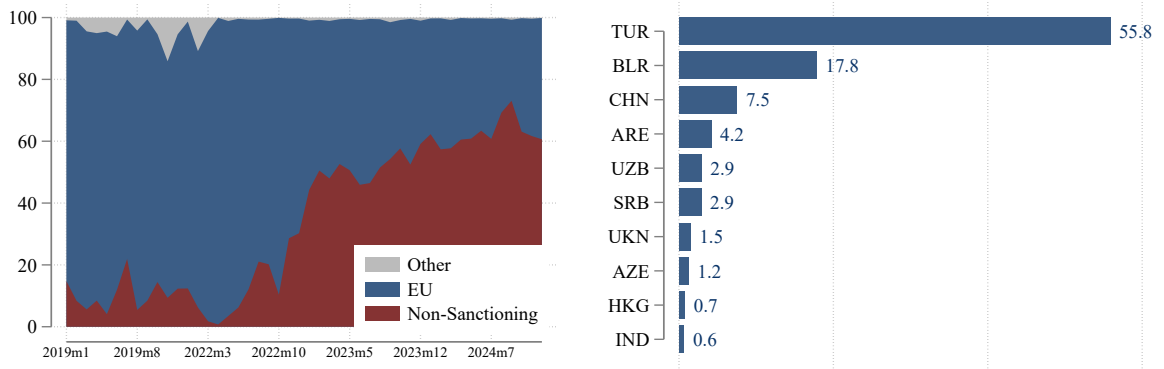
FIGURE 4. RUSSIAN IMPORTS OF EU-ORIGIN GOODS SUBJECT TO EXPORT BANS

Note: This figure plots the monthly value (billion USD) and quantity (kilotonnes) of Russian imports from the EU of products that later became subject to EU full export bans. Data are available for 2019 and for 2022–2024. Dashed vertical lines indicate Russia’s invasion of Ukraine in February 2022 and the introduction of the EU export bans in July 2022. After the export bans take effect, any remaining imports of these export-banned EU-origin products reflect evasion of the bans. We exclude products in HS2 88 (aircraft and parts) as they are extreme outliers.

Fact 2: Evasion occurs through distinct transshipment modes We use information on the country of dispatch to distinguish between different transshipment modes in the data. Figure 5a plots the share of Russian imports of EU-origin, export-banned products by dispatch country over time. Prior to the introduction of the export bans, the average monthly share of EU-origin shipments dispatched from EU countries is 87.4%, reflecting standard export patterns in which goods are shipped directly from the EU or routed through other EU countries before reaching Russia. Immediately after the export bans enter into force in July 2022, most evading shipments continue to be dispatched

from EU countries. In this initial phase, evasion therefore predominantly occurs through transactions in which goods retain both an EU origin and an EU dispatch country, consistent with in-transit transshipment, where goods are redirected to Russia during shipping without a change in the exporter of record. Over time, however, the composition of transshipment modes shifts toward re-exporting. Starting after the export bans enter into force, the share of EU-origin imports dispatched from non-EU countries rises steadily to 60.6% by the end of 2024, with the steepest increase occurring between July 2022 and February 2023, when this share more than quadruples from 12.1% to about 50.5%.

Three patterns emerge from the evolution of transshipment modes. First, in-transit transshipment is particularly prevalent immediately after the export bans enter into force, when existing shipping arrangements continue to be used. Second, re-exporting expands more gradually but steadily over time, eventually becoming the dominant mode of evasion, consistent with the presence of fixed costs involved in building up this mode of transshipment. Third, while re-exporting continues to grow after early 2023, its expansion slows markedly. This timing coincides with a period in which the EU and its partners intensified efforts to curb transshipment starting in March 2023, suggesting that stricter enforcement disproportionately affects the growth of re-exporting.



(A) IN-TRANSIT VS. RE-EXPORT TRANSSHIPMENT

(B) TOP 10 HUBS OF RE-EXPORTING

FIGURE 5. TRANSSHIPMENT MODES AND RE-EXPORTING HUBS

Note: Panel (A) shows the share of Russian imports of EU-origin, export-banned products by dispatch country over time, distinguishing in-transit transshipment (EU dispatch) from re-exporting transshipment (non-sanctioning dispatch). Panel (B) reports the top ten re-exporting hubs by their share of total re-exported trade. Hub names are reported as ISO-3 country codes; UKN denotes unknown.

Fact 3: Re-exporting is highly concentrated in a few hubs Re-exporting of EU-origin, export-banned products is highly concentrated geographically. Figure 5b shows the distribution of re-exported trade across intermediary countries. Three hubs—Türkiye, Belarus, and China—account for the majority of re-exporting activity. Türkiye alone accounts for more than half of total re-exported trade, while Belarus (17.8%) and China

(7.5%) contribute substantial additional shares. At the same time, re-exporting is not confined to these three countries. An additional 18.9% of re-exported trade is routed through a broader set of intermediary hubs. Within the set of remaining intermediary hubs, re-exporting is itself concentrated: the United Arab Emirates, Uzbekistan, and Serbia together account for around 60% of re-exported trade routed through these hubs. The remaining share is distributed more evenly across a set of smaller intermediaries, including Azerbaijan, Hong Kong, India, Kyrgyzstan, Turkmenistan, Georgia, and Kazakhstan. Evasion through re-exporting therefore relies heavily on a small number of key hubs, while also exploiting a wider network of alternative routes.

Fact 4: Evasion varies widely across products Figure 6 illustrates substantial heterogeneity in evasion across products affected by the export bans. Panels (A) and (B) plot the cumulative distribution of product-level evasion across the 688 fully export-banned HS6 products (by July 2022), where evasion is measured as the ratio of average monthly EU-origin imports in the post-bans period (August 2022–December 2024) to the average monthly imports in 2019. Values close to one reflect post-ban imports near their pre-war levels, values above one reflect imports exceeding their 2019 average, and values close to zero reflect a collapse in imports following the bans. Panel (A) reports the distribution using import values, while Panel (B) uses quantities. At the lower end of the distribution, 9.7% of products are no longer imported following the introduction of export bans, while a further 24.5% exhibit evasion rates below 20%, indicating that imports of these products decline to only a small fraction of their pre-war levels. At the upper end, 37.3% of products have evasion rates of at least 80%, including 29% with evasion rates above 100%, for which post-ban imports exceed their 2019 averages. A similar picture emerges for quantities.

We next relate this heterogeneity to pre-war product characteristics measured in 2019. Panels (C)–(F) of Figure 6 show binned scatter plots of product-level evasion against a set of characteristics that proxy for the ease with which shipments can be concealed within normal trade flows and for the availability of substitutes outside the EU. These characteristics include measures of market concentration on the supplier and buyer sides, indicators of prior third-country routing, and physical attributes related to bulkiness, proxied by product weight.

Evasion is higher for products characterized by less concentrated supplier markets, indicating the presence of a larger set of alternative suppliers, and for products purchased by a less concentrated set of Russian buyers which indicates that many trade relationships are already established. This pattern is consistent with evasion being easier for products that circulate in larger volumes through these existing networks, as such flows both provide cover within legitimate trade and benefit from firms’ existing know-how in handling the products, including familiarity with documentation, packaging, and customs procedures.

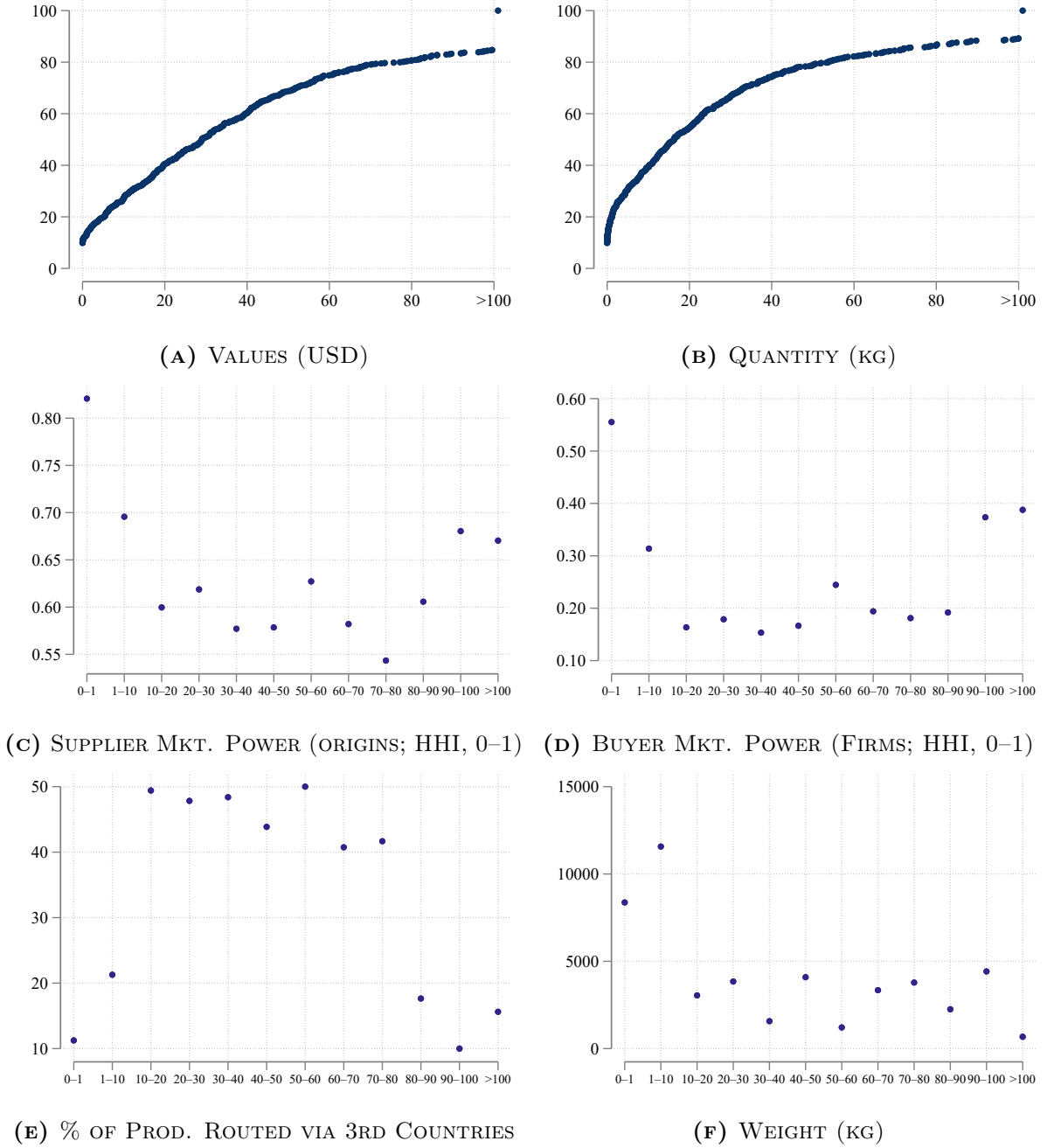


FIGURE 6. HETEROGENEITY IN EVASION ACROSS PRODUCTS

Note: Panels (A) and (B) plot the cumulative distribution of evasion across the 668 HS6 products subject to EU export bans introduced as of July 2022. Evasion is measured as the ratio of average monthly Russian imports of EU-origin products in the post-ban period (August 2022–December 2024) to the corresponding average in 2019; values above 100% indicate post-ban exports that exceed their 2019 levels. Panel (A) constructs the cumulative distribution of evasion using import values (USD), while Panel (B) uses quantities (kg). Panels (C)–(F) relate product-level evasion to pre-war product characteristics measured in 2019. In Panel (C), supplier market power is measured using the Herfindahl–Hirschman Index (HHI), computed across the product’s countries of origin. In Panel (D) buyer market power is measured using the Herfindahl–Hirschman Index (HHI), computed across the product’s importing firms. Each point reports the unweighted mean of the characteristic shown on the y-axis within a bin of the evasion measure, except for weight, which are reported as medians within bins.

Evasion is also higher for products that were already routed through third countries prior to the bans, even at low levels, suggesting that existing third-country routing facilitates rerouting. This pattern is consistent with the presence of sunk costs in establishing transshipment routes: even limited pre-war use of intermediaries appears to lower the barriers to expanding indirect trade after the bans. In addition, evasion is systematically higher for lighter products, which are cheaper to divert and easier to conceal within normal trade flows. By contrast, unit values do not exhibit a systematic relationship with evasion, nor does the typical bundling of multiple HS6 products in shipments, which serves as a proxy for the ease of concealing goods within mixed consignments (Appendix Figure A4).

Fact 5: The most heavily evaded products are hard to substitute A distinct pattern emerges for products in the extreme right tail of the evasion distribution, namely those for which post-ban imports exceed their 2019 levels. This group is quantitatively important: 29.0% of products in value terms and 18.3% in quantity terms fall into this category. Relative to other export-banned products, these top-evaded goods are characterized by higher concentration in the supplier market, as measured by an HHI computed across countries with the EU treated as a single supplier, and by a more concentrated set of Russian buyers, consistent with limited scope for substitution. At the same time, they exhibit moderate levels of pre-war third-country routing: on average, 25.6% of these products had at least one shipment in 2019 that was routed from the EU through a non-coalition intermediary before reaching Russia, a share similar to that of products with evasion rates below 20%. By contrast, products with high (though not above-2019 import-level) evasion rates exhibit substantially higher pre-war routing shares. For products with evasion rates between 70–90%, average routing shares range from 48.3–50.0%, before declining slightly to 46.4% for products with evasion rates between 90–100%.

Taken together, these patterns indicate that the most heavily evaded products combine high market concentration on the supplier and buyer sides with few pre-existing transshipment channels. For this subset of goods, continued access appears particularly important, so that Russia’s demand is met through especially intensive evasion, despite higher costs. This contrasts with the larger set of export-banned products, which highlights that evasion not only reflects concealment opportunities but also the difficulty of substitution.

These facts naturally map into the structure of the quantitative model we develop below. The model (i) features endogenous transshipment through third-country hubs, (ii) is calibrated to the 6-digit HS level so that trade elasticities and trade costs—and, hence, the transshipment patterns—vary by product, and (iii) will match Russia’s pre-war product-level import structure, allowing substitution patterns and evasion intensity to differ across products.

4 A Quantitative Model of International Trade and Transshipment

We develop a general-equilibrium model of international trade with endogenous transshipment. When evaluating export bans, the framework distinguishes between evasion and trade diversion (i.e., substitution with non-sanctioning suppliers). This allows us to quantify the aggregate consequences of sanctions, to understand the mechanisms behind evasion, and to compute different counterfactual scenarios, such as full enforcement or closing specific transshipment hubs.

4.1 The Framework

The framework builds on the multi-product version of [Eaton and Kortum \(2002\)](#) (henceforth, EK), similar to [French \(2016\)](#). Countries trade intermediate varieties that are used to compose a finite set of final consumption goods. Countries are indexed by $e, i = \{1, 2, \dots, N\}$, where e denotes the exporter and i the importer, while final goods are indexed by $k = \{1, 2, \dots, K\}$. We use “goods” to refer to the final-consumption-goods level, k , while “varieties” are the tradable intermediate inputs that compose the final goods. Households consume the final goods, assembled using product-specific tradable intermediate varieties. They inelastically supply labor to competitive domestic intermediate producers, which is the only production input. In what follows, we detail each of these layers.

4.1.1 Demand side: Households

There are L_i households in country i that use product-specific tradable intermediate inputs to assemble the final consumption goods.⁸ These products are non-tradable, and prices are taken as given, both of which are standard assumptions in the literature.

The representative household in country i inelastically supplies one unit of labor to domestic producers and has standard CES utility over a finite set of non-tradable consumption goods

$$U_i = \left[\sum_{k=1}^K (q_i^k)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1.$$

⁸ Alternatively, we could add a competitive final-goods sector that assembles the non-tradable final products and sells to the households, which would deliver the same results.

Every final good k is composed of a continuum of k -specific tradable varieties, with the measure of varieties normalized to one,

$$q_i^k = \left[\int_0^1 (q_i^k(\omega^k))^{\frac{\eta^k-1}{\eta^k}} d\omega^k \right]^{\frac{\eta^k}{\eta^k-1}}, \quad \eta^k > \sigma.$$

We set $\eta^k > \sigma$ to account for the fact that varieties within the same category are closer substitutes than products across categories, as in French (2016). Note that the assembly function can also be seen as a lower-tier utility function.

The household maximizes its utility subject to its budget constraint, $E_i \geq \sum_{k=1}^K \left[\int_0^1 p_i^k(\omega^k) q_i^k(\omega^k) d\omega^k \right]$, where E_i is its total income, and $p_i^k(\omega^k)$ the price of variety ω^k , which is the minimum price from sourcing around the globe, as explained in the next subsection. Cost minimization of assembling q_i^k yields the k -level price index, $P_i^k = [\int_0^1 (p_i^k(\omega^k))^{1-\eta^k} d\omega^k]^{1/(1-\eta^k)}$, and the aggregate price index is given by

$$P_i^{1-\sigma} = \sum_{k=1}^K (P_i^k)^{1-\sigma}, \quad (1)$$

and the expenditure shares are as follows

$$\frac{P_i^k q_i^k}{\sum_{l=1}^K P_i^l q_i^l} = \left(\frac{P_i^k}{P_i} \right)^{1-\sigma}.$$

Next, we outline how the intermediate varieties are produced and shipped.

4.1.2 Supply Side: Production of Tradable Varieties

Intermediate varieties are produced and shipped by perfectly competitive firms using labor as the only input in production.

Trade costs. Throughout the paper, we index the exporter by e and the importer by i . A subscript ei can be read as “from country e to i .” International trade is costly: intermediate inputs are traded subject to standard iceberg trade costs. Firms in e need to ship $\tau_{ei}^k \geq 1$ units of a variety ω^k for one unit to arrive in i . We normalize $\tau_{ii}^k = 1$ and assume that the so-called triangle inequality holds, such that $\tau_{ei}^k \leq \tau_{eh}^k \tau_{hi}^k \forall (e, h, i)$. This no-arbitrage condition will be central in the transshipment analysis, as detailed in Section 4.2.

Production technology. Labor is perfectly mobile across industries but immobile across countries, which yields a common wage across industries. Firms operate in perfect

competition with constant returns to scale, and let $\varphi_e^k(\omega^k)$ denote the labor productivity in country e to produce variety ω^k . Hence, the cost of producing one unit of ω^k and shipping it to i is given by the marginal cost, $c_{ei}^k(\omega^k) := (w_e \tau_{ei}^k) / \varphi_e^k(\omega^k)$, where w_e is the wage rate in country e . Note that $c_{ei}^k(\omega^k)$ is also the price offered by e to i under perfect competition.

Following the literature, $\varphi_e^k(\omega^k)$ is independently drawn from a Fréchet distribution, $F_e^k(\varphi) = \exp(-T_e^k \varphi^{-\theta^k})$, for $\varphi > 0$. T_e^k measures the overall technology level in a country e to produce the varieties for product k , while θ^k governs the forces of comparative advantage across varieties of a product, as it is inversely related to the dispersion of productivities.

4.1.3 Trade Flows and Equilibrium

Trade. Consumers pick the lowest-cost source of a variety across the world. This, coupled with the Fréchet distribution, yields the familiar gravity equation of trade flows (i.e., the expenditure share in country i on varieties from e and product k)⁹

$$\pi_{ei}^k := \frac{X_{ei}^k}{X_i^k} = \frac{T_e^k (w_e \tau_{ei}^k)^{-\theta^k}}{\sum_{l=1}^N T_l^k (w_l \tau_{li}^k)^{-\theta^k}}, \quad (2)$$

where X_{ei}^k is the expenditure level of i on product- k varieties from e , and $X_i^k := \sum_{e=1}^N X_{ei}^k$ is i 's total expenditure. We refer to π_{ei}^k as the “trade share.” Intuitively, trade flows are higher, the more productive e is on average (higher T_e^k), and the lower its production and transportation costs (lower $w_e \tau_{ei}^k$) relative to the other countries. We typically refer to θ^k as the “trade elasticity,” because a higher θ^k implies that trade flows are more elastic with respect to changes in trade costs. The denominator in equation (2) determines the price index, that is,¹⁰

$$P_i^k = \left[\sum_{e=1}^N T_e^k (w_e \tau_{ei}^k)^{-\theta^k} \right]^{-1/\theta^k}. \quad (3)$$

Equilibrium. Balanced trade closes the model and determines equilibrium wages

$$\sum_{e \neq i} \sum_{k=1}^K (\pi_{ei}^k X_i^k - \pi_{ie}^k X_e^k) = 0 \quad \forall i. \quad (4)$$

⁹ This follows from standard derivations (see [Costinot et al., 2012](#); [Eaton and Kortum, 2002](#)).

¹⁰ Technically, we obtain $P_i^k = \Gamma \left(\frac{\theta^k + 1 - \eta^k}{\theta^k} \right)^{1/(1-\eta^k)} \left[\sum_{e=1}^N T_e^k (w_e \tau_{ei}^k)^{-\theta^k} \right]^{-1/\theta^k}$. As the Gamma function, $\Gamma(\cdot)$, is a constant that adds notation but has no bearing on our estimations or quantifications, we will ignore it henceforth. One could alternatively simply add an appropriate k -specific scalar to each T_e^k .

As labor is the only source of household income, we have $E_i = w_i$, and in equilibrium we must have $X_i = w_i L_i$, where $X_i := \sum_k X_i^k$ denotes aggregate expenditure. Accordingly, using households' expenditure shares from above, we obtain

$$\frac{X_i^k}{X_i} = \left(\frac{P_i^k}{P_i} \right)^{1-\sigma}. \quad (5)$$

4.1.4 Solving the Model in Changes

Following a shock to trade costs, we can solve the model in relative changes for a given set of elasticities and baseline trade flows (including domestic expenditure). This procedure is often called “exact hat algebra” and was popularized by [Dekle et al. \(2007\)](#).

Let x' denote the counterfactual value of variable x after the shock, and denote the relative changes by $\hat{x} := x'/x$. Denote total bilateral trade flows by $X_{ei} := \sum_k X_{ei}^k$. Using equations (1)–(5), it follows that the equilibrium in relative changes after a change in τ is given by

$$\hat{P}_i^k = \left[\sum_{e=1}^N (\hat{w}_e \hat{\tau}_{ei}^k)^{-\theta^k} \pi_{ei}^k \right]^{-1/\theta^k} \quad (6)$$

$$\hat{P}_i = \left[\sum_{k=1}^K (\hat{P}_i^k)^{1-\sigma} \frac{X_i^k}{X_i} \right]^{1/(1-\sigma)} \quad (7)$$

$$(\pi_{ei}^k)' = \frac{(\hat{w}_e \hat{\tau}_{ei}^k)^{-\theta^k} \pi_{ei}^k}{\sum_{l=1}^N (\hat{w}_l \hat{\tau}_{li}^k)^{-\theta^k} \pi_{li}^k} \quad (8)$$

$$(\pi_{ei})' = \sum_{k=1}^K (\pi_{ei}^k)' \left(\frac{\hat{P}_i^k}{\hat{P}_i} \right)^{1-\sigma} \frac{X_i^k}{X_i} \quad (9)$$

$$\hat{w}_i X_i = \sum_{e=1}^N (\pi_{ie})' \hat{w}_e X_e. \quad (10)$$

The equilibrium consists of a set of wage changes that, in turn, determine the changes in price indices and trade flows. We solve for this set of wages using a tâtonnement algorithm (see also [Alvarez and Lucas, 2007](#)). In what follows, we detail how transshipment can happen in the model, before explaining how we bring the model to the data.

4.2 Transshipment

To measure transshipment opportunities, we rely on the triangle inequality (TI) of trade costs, $\tau_{ei}^k \leq \tau_{eh}^k \tau_{hi}^k \forall (e, h, i)$, following recent insights from [Foellmi et al. \(2024\)](#)—for more details, consult the paper. In words, the TI is a no-arbitrage condition, stating that

the costs we assign to shipments from e to i cannot exceed the costs of shipping a good from e to h and then re-shipping it to i . Otherwise, if we had $\tau_{ei}^k > \tau_{eh}^k \tau_{hi}^k$, firms in e would start using h as an intermediary, and the actual trade cost from e to i becomes $\tau_{ei}^k = \tau_{eh}^k \tau_{hi}^k$. This can be microfounded with link-level trade costs (Foellmi et al., 2024), and accordingly, the TI does not imply that all trade goes directly from e to i , but simply that the bilateral trade cost τ_{ei}^k represents the minimum cost to ship from e to i , be it directly or indirectly. For notational convenience, we suppress the product superscript k in what follows whenever this does not cause confusion.

The crucial insight is that the TI has implications for counterfactual exercises. After a shock to trade costs, we need to verify the TIs for each combination of (e, h, i) . If the TI is violated for a triplet, we update the trade costs. For instance, suppose trade costs decrease uniformly to some fraction $\delta < 1$ of their baseline values. If $\delta\tau_{ei} > (\delta\tau_{eh})(\delta\tau_{hi})$ for some h , we set the effective $e \rightarrow i$ trade cost equal to $(\delta\tau_{eh})(\delta\tau_{hi})$. The updating is done iteratively, which verifies whether multi-hub paths further reduce trade costs, such that in the end we have $\tau'_{ei} = \min_h \{\tau'_{eh} \tau'_{hi}\}$.¹¹ In other words, once trade costs change, we recompute bilateral trade costs as the cost of the cheapest available “route” between any two countries, so transshipment arises whenever re-exports are cheaper than direct exports.

A consequence of the TI logic is that we need to calibrate the initial level of trade costs to verify transshipment opportunities (i.e., TI violations) after a shock. Without knowing the level of trade costs, it is not possible to evaluate which country is the best hub. For instance, in our example with a common δ , knowing solely the size of δ reveals nothing about whether $\delta\tau_{ei} > (\delta\tau_{eh})(\delta\tau_{hi})$ holds. Therefore, after a shock, we use the initial level of calibrated trade costs to ensure that in the counterfactual scenario we have $\tau'_{ei} = \min_h \{\tau'_{eh} \tau'_{hi}\}$. Once we know the counterfactual value of trade costs, τ'_{ei} , we set $\hat{\tau}_{ei} = \tau'_{ei} / \tau_{ei}$ and solve the model in relative changes, as detailed in Section 4.1.4.

In our application in the context of export bans, an EU source e will select, for each product k , the hub h that minimizes the indirect costs to Russia, as depicted in Figure 7. In counterfactual scenarios where we do not allow transshipment, we impose a prohibitive cost on EU exports to Russia and do not verify the TI after the shock, as is the typical procedure in the literature.

¹¹ Foellmi et al. (2024) show that this leads to a unique solution after a finite number of iterations.

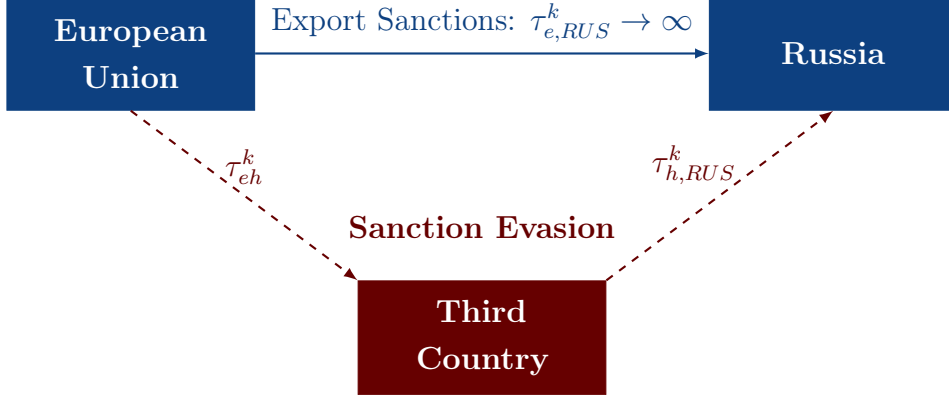


FIGURE 7. AN ILLUSTRATION OF TRANSSHIPMENT VIA THE TRIANGLE INEQUALITY

Notes. This figure depicts the mechanism of transshipment in our model, i.e., the triangle inequality. When the direct trade costs from an EU source e to Russia become prohibitive, a third country h is used to re-export the varieties of product k , which entails a cost equal to $\tau_{eh}^k \tau_{h,RUS}^k$.

Note that our approach matches the structure of our data. In the customs data, we observe the origin country, the shipping (“dispatch”) country, and the destination (Russia). This is precisely what we can contrast with the model, with e being the origin, h being the shipping country, and i being Russia. We do not model the full underlying transportation network, like in the frameworks of [Allen and Arkolakis \(2022\)](#) and [Ganapati et al. \(2024\)](#), because we neither observe nor aim to explain the specific path taken to reach, say, h from e . Moreover, in the model, each origin-product cell uses a single least-cost transshipment hub. This feature is consistent with Russian customs data: for EU origins and fully banned products with positive imports in 2024, about half of origin-product cells use exactly one hub (i.e., dispatch country), and among those using multiple hubs, the median share routed through the main hub is as high as 76%.

Before we outline how to bring the model to the data, three further remarks about the TI are in order. First, in this setting, the hub has no rent from transshipment, because the transportation sector is perfectly competitive. Second, as trade costs are deterministic, only the least-cost intermediary is used in equilibrium. Third, the TI’s form implies that there are, on the one hand, no additional costs associated with connecting multiple exporters, while, on the other hand, a transshipment flow via a hub h entails the full costs of exporting from e to h and then from h to i . While such features could be refined, they arise from the basic structure of canonical trade models, and we will show that, despite its simplicity, our setup has considerable predictive power over which hubs are used in the Russian customs data.

4.3 Bringing the Model to the Data

Section 4.1.4 shows that we need to know (i) initial trade flows, X_{ei}^k , and (ii) a set of elasticities, θ^k and σ , to compute a counterfactual equilibrium for a given change in τ . In addition, to implement TI-based transshipment, we require a baseline level of bilateral trade costs, τ_{ei}^k , as explained in Section 4.2. In this section, we also identify the coalition group of countries imposing export bans.

We calibrate the model so that each product k corresponds to a 6-digit HS code, the most disaggregated level available that is harmonized across countries. This disaggregation level comes with a data challenge, as explained below, but allows us to overcome several drawbacks of more aggregated quantitative exercises that are standard in the literature. First, we can impose export bans directly at the product level, whereas a sector-level quantification would require an ad hoc approach to aggregating the shock, since not all products in a sector are banned. Second, the model then predicts transshipment patterns at the origin-product level, which we can compare with our disaggregated customs data. Third, this introduces product-specific trade dependencies and elasticities to better approximate the substitutability of an origin-product cell.

Data. We use data from 2019, the last full year before COVID and the war. Bilateral product-level trade flows from UN Comtrade measure $X_{ei}^k \forall (e \neq i)$, with a product defined as a 6-digit HS code in the HS2017 vintage. For each flow, we use the maximum between the importer’s recorded import and the exporter’s reported export. To add total expenditure, X_i , we take the OECD Inter-Country Input-Output tables ([regular ICIO](#)). For countries missing in the ICIO tables, we use the International Trade and Production Database for Estimation ([ITPD-E \(Release 2\)](#); see [Borchert et al. \(2021\)](#)). The ITPD-E data sometimes yield very small (and even negative) domestic expenditure. We drop countries with unreasonably low domestic shares below 10%. Moreover, to reduce the dimensionality of the quantification, given that we have over 5,000 products in the final sample, we keep a subset of countries from the UN Comtrade data. We restrict the sample to countries that are either (i) in Europe, (ii) in the coalition, (iii) sharing a border with Russia, (iv) in the Eurasian Economic Union, (v) typical transshipment hubs (like UAE and Thailand), or (vi) among Russia’s top 60 source countries. Moreover, we exclude very small countries with a total population below two million. This leaves a total of 74 countries, which make up 96% (98%) of Russia’s total exports (imports).¹² Finally, we drop HS6 products for which Russia’s total 2019 trade is less than USD 10,000 both on the import and on the export side, which leaves 5136 products in the final sample.

¹² We treat China and Hong Kong as a single country when calibrating the model (reassigning Hong Kong’s flows to China) due to large shares of re-exports between these countries, as is frequently done in the quantitative trade literature.

As is typical in the literature, the elasticity values are taken off the shelf. We take θ^k from [Fontangé et al. \(2022\)](#), who estimate product-level trade elasticities using gravity regressions consistent with our setup above. For 6-digit products that cannot be matched to their data, we use the average elasticity within the product’s 4-digit industry. Moreover, due to some outliers in the elasticity data, we winsorize by bounding values below the first and above the 99th percentiles. The resulting minimum, median, and maximum of θ^k are, respectively, 1.5, 8.2, and 44.1. We set $\sigma = 2.2$, in line with [French \(2016\)](#).

We cannot read X_i^k (i.e., product-level expenditure) directly from the data, because expenditure is not observable at this level of disaggregation globally. We therefore recover k -level expenditure—and, accordingly, domestic product-level trade—using the model’s structure.¹³

Product-level expenditure. From equation (5) we know that if we observe P_i^k and X_i , we can deduce X_i^k . We observe X_i directly from the data, but need to estimate P_i^k . We cannot simply use equation (2) to estimate P_i^k , because the dependent variable is scaled by X_i^k . However, using equations (1), (2), (3), and (5), we obtain

$$\frac{X_{ei}^k}{X_{ei}} = \frac{T_e^k w_e^{-\theta^k} (P_i^k)^{\theta^k - (\sigma - 1)} (\tau_{ei}^k)^{-\theta^k}}{\sum_{s=1}^K T_e^s w_e^{-\theta^s} (P_i^s)^{\theta^s - (\sigma - 1)} (\tau_{ei}^s)^{-\theta^s}}.$$

Taking logs of this equation yields

$$\log \left(\frac{X_{ei}^k}{X_{ei}} \right) = \log \left(T_e^k w_e^{-\theta^k} \right) + \log \left((P_i^k)^{\theta^k - (\sigma - 1)} \right) - \log (\Psi_{ei}) - \theta^k \log (\tau_{ei}^k),$$

where $\Psi_{ei} := \sum_{s=1}^K T_e^s w_e^{-\theta^s} (P_i^s)^{\theta^s - (\sigma - 1)} (\tau_{ei}^s)^{-\theta^s}$. We run a regression with $\log (X_{ei}^k / X_{ei})$ as the dependent variable. The first three elements on the right-hand side can be captured by an exporter-product fixed effect (FE), an importer-product FE, and an exporter-importer FE, respectively. To parametrize the impact of τ_{ei}^k , we interact a set of product dummies with several standard bilateral variables: the log of population-weighted distance, as well as dummy variables indicating whether the country pair has a free trade agreement or is part of the same customs union, and whether it shares a common border, language, or colonial history. These control variables are obtained from the standard [CEPII Gravity database](#) ([Conte et al., 2022](#)).

Therefore, through the lens of the model, the exponentiated importer-product FE yields $(P_i^k)^{\theta^k - (\sigma - 1)}$. Using our values for θ^k and σ described above, we obtain an estimate of P_i^k . For countries where we cannot estimate this price index, we set it equal to the maximum

¹³ [French \(2016\)](#) recovers product-level expenditure while assuming that trade costs and trade elasticities are common across products, whereas [Torun \(2025\)](#) uses a non-homothetic final-goods aggregator to do so.

estimate within k (recall that a higher P_i^k implies more difficult access to cheap sources). Moreover, due to outliers in these estimates, we winsorize P_i^k by bounding values below the 10th and above the 90th percentiles within each importer i . Equations (1) and (5) then yield X_i^k , with which we can calculate the trade shares, π_{ei}^k . As a final step, because the home share, together with the elasticities, is known to be the main determinant of the welfare changes in these types of models, we set X_i^k such that the home shares, π_{ii}^k , never fall below 10% to avoid obtaining extreme results.

Trade costs. To solve the model as described in Section 4.1.4 is the change in trade costs, $\hat{\tau}$. When we block transshipment in the counterfactual scenario, we simply set $\hat{\tau}_{e,RUS}^k \rightarrow \infty$ for all e - k combinations where we observe an export ban.

However, when allowing for transshipment, we need to know the initial trade-cost level to verify the TIs after the shock, as explained in Section 4.2. To recover trade barriers, we use the trade shares, π_{ei}^k , and the estimated exporter-specific terms $T_e^k w_e^{-\theta^k}$. Similar to the above, for countries where we cannot estimate $T_e^k w_e^{-\theta^k}$, we set this equal to the minimum estimate within k (recall that a lower $T_e^k w_e^{-\theta^k}$ implies less export competitiveness), and we winsorize by bounding values below the 10th and above the 90th percentiles within each exporter e . According to the gravity equation, trade barriers can be calculated using bilateral relative to domestic flows, that is,

$$\tau_{ei}^k = \left(\frac{\pi_{ei}^k / (T_e^k w_e^{-\theta^k})}{\pi_{ii}^k / (T_i^k w_i^{-\theta^k})} \right)^{-1/\theta^k}, \quad (11)$$

which will match trade flows exactly, in line with the hat-algebra logic.

As a final step, we need to ensure that the initial trade costs satisfy the TIs to have a theory-consistent starting point. Estimates of trade barriers can violate the TI, which can be explained with, e.g., the presence of bilateral preference shifters (Foellmi et al., 2024). To recover trade costs, we apply the algorithm from Foellmi et al. (2024, Section 4) and update the τ_{ei}^k values from equation (11) iteratively until we have no more TI violations. This yields a set of TI-consistent trade costs, which we then use to measure transshipment opportunities following a shock.¹⁴

Coalition group. In the counterfactual exercises below, we need to specify which countries sanctioned (banned) which products. These countries will form the “coalition.”

¹⁴ The algorithm starts by setting the trade barriers that we obtain from equation (11) to at least one. Then, it verifies the TIs, and if a TI is violated, it sets the updated trade cost equal to the minimum indirect cost. This procedure is repeated until there are no more TI violations. The wedge between the initial value of trade barriers and the final value of trade costs can be interpreted as a bilateral preference shifter through the lens of standard gravity theory (cf. Foellmi et al., 2024).

Following Li et al. (2024), we identify 38 U.S.-aligned countries exempt from U.S. licensing requirements for exports to Russia, as listed in Supplement No. 3 of Part 746 of the Export Administration Regulations (EAR), which recognizes countries implementing “substantially similar export controls.” Beyond the EU-27, several countries align with EU sanctions to varying degrees, through selective adoption, voluntary compliance, or formal implementation as part of EU accession. A core group—including Albania, Bosnia and Herzegovina, Iceland, Liechtenstein, Montenegro, North Macedonia, Norway, and Ukraine—consistently follows EU sanction packages.¹⁵ For these countries, we assume the EU sanction regime applies identically. For the remaining countries (see part III of Table 1), we rely on product-level sanction information from the Global Trade Alert (GTA) database. We retain sanctions at the 6-digit product level, applied to either all firms or individual firms, classified as discriminatory, and with an implementation date before December 2024. In the absence of explicit entry-into-force dates, we define sanctions using a two-step approach: (i) product–country observations are explicitly listed as sanctioned in GTA, and (ii) product–country–time observations for which exports to Russia fall below 10% of their pre-shock average.¹⁶ The EU’s sanction regime broadly encompasses the products targeted by these coalition countries (Table 2), justifying their designation as such. Nevertheless, with roughly 3,000 sanctioned products, the EU’s coverage extends beyond that of its coalition partners. For non-EU coalition countries, GTA provides product-level sanction flags but does not distinguish between partial and full bans across 6-digit codes. We therefore refrain from using these countries in analyses that require a precise full-versus-partial classification, such as in the triple-difference event-study design of Section 5.

The EU has also extended its sanction regime to Belarus in response to the country’s involvement in Russia’s military aggression against Ukraine. This is important for our quantification, because we will block transshipment via Belarus in the model for the products where it was also hit by sanctions. Sanction data are drawn from TARIC, applying the same inheritance rules as for EU-Russia sanctions. By July 2022, 1,195 products were subject to export bans, rising to 1,299 products by December 2024 (see Figure A1). The overlap with EU-Russia sanctioned products was initially limited: only 29% of products fully sanctioned against Russia by July 2022 were also fully sanctioned vis-à-vis Belarus. By December 2024, this share increased to approximately 38%.¹⁷ The

¹⁵ <https://www.consilium.europa.eu/en/press/press-releases/2025/12/08/statement-by-the-high-representative-on-behalf-of-the-eu-on-the-alignment-of-certain-countries-concerning-restrictive-measures/>

¹⁶ For product–origin pairs absent from the pre-period, we proxy the EIF date with the implementation date.

¹⁷ For EU–Belarus sanctions, the implementation date rather than the entry-into-force date is used to define sanctioned products. Consequently, these overlap figures likely represent an upper bound, as wind-down periods and exemptions may delay the actual enforcement of the sanctions.

TABLE 1. SANCTIONING COALITION COUNTRIES

EU-27			
AUT — Austria	BEL — Belgium	BGR — Bulgaria	CYP — Cyprus
CZE — Czechia	DEU — Germany	DNK — Denmark	ESP — Spain
EST — Estonia	FIN — Finland	FRA — France	GRC — Greece
HRV — Croatia	HUN — Hungary	IRL — Ireland	ITA — Italy
LTU — Lithuania	LUX — Luxembourg	LVA — Latvia	MLT — Malta
NLD — Netherlands	POL — Poland	PRT — Portugal	ROU — Romania
SVK — Slovakia	SVN — Slovenia	SWE — Sweden	—
Mirrors EU sanctions			
ALB — Albania	BIH — Bosnia & Herzegovina	ISL — Iceland	LIE — Liechtenstein
MKD — North Macedonia	MNE — Montenegro	NOR — Norway	UKR — Ukraine
Other sanctioning countries			
AUS — Australia	CAN — Canada	CHE — Switzerland	GBR — UK
JPN — Japan	KOR — South Korea	NZL — New Zealand	SGP — Singapore
TWN — Taiwan	USA — United States	—	—

Note: The table distinguishes three groups of sanctioning countries. The first group comprises the EU-27 member states. The second group includes non-EU countries that closely mirror the EU sanctions regime and for which the same sanctions are applied consistently. The third group consists of non-EU coalition countries that have imposed their own sanctions regimes on Russia, which differ to varying degrees from the EU regime.

TABLE 2. COVERAGE OF EU-RUSSIA SANCTIONS BY NON-EU COALITION COUNTRIES

Country	Share of products covered by EU's regime [%]	Total products sanctioned	Overlap with EU
Singapore	100.0	15	15
Switzerland	99.8	432	431
Canada	98.6	148	146
South Korea	97.7	386	377
United Kingdom	96.9	294	285
USA	88.9	416	370
Japan	74.2	458	340
Australia	70.0	30	21
New Zealand	42.9	7	3

Note: This table reports, for each country in the non-EU coalition group, the extent to which its sanctions regime with respect to Russia is covered by the EU-Russian sanction regime, measured as the share of sanctioned products that are also sanctioned by the EU vis-a-vis Russia. Columns (2) and (3) report the total number of products sanctioned by each country and the number of those products that are also sanctioned by the EU.

sanctioned products for the remaining coalition countries are again taken from Global Trade Alert (GTA).

5 Estimating the Actual Trade-Cost Increase

To quantify the de facto increase in trade costs implied by the export bans, we bring the model's gravity structure to the data. The object of interest is the change in bilateral trade costs for fully sanctioned EU products when exporting to Russia. We estimate this using a triple-difference (DDD) event-study design at the HS6–origin–month level for 2019 and 2022–2024. We restrict the sample to products that were either never sanctioned or fully banned by July 2022, when the first major wave of export bans became binding. This avoids complications due to staggered treatments and ensures a clean comparison

between treatment and control observations. By excluding partly sanctioned products, we can cleanly identify actual evasion, rather than potentially misinterpreting trade in non-sanctioned product subgroups as such.

The estimating equation follows directly from equation (2), where export bans enter as an increase in iceberg trade costs for banned products. Combining $X_{e,RUS}^k = \pi_{e,RUS}^k X_{RUS}^k$ with equation (2) and taking logs yields,

$$\log(X_{e,RUS}^k) = \underbrace{\log(T_e^k w_e^{-\theta^k})}_{\text{exporter-specific}} + \underbrace{\log(X_{RUS}^k (P_{RUS}^k)^{\theta^k})}_{\text{importer-specific}} - \underbrace{\theta^k \log(\tau_{e,RUS}^k)}_{\text{bilateral trade cost}}, \quad (12)$$

where the first and second elements on the right-hand side are exporter- and importer-specific, respectively. The only bilateral component left is the trade cost, whose net increase is precisely what we want to measure. In other words, the gravity theory guides us in setting up exporter- and importer-specific controls to isolate trade-cost changes, such that the treatment coefficients below identify changes in $-\theta^k \log(\tau_{e,RUS}^k)$. To then translate the increase in $\tau_{e,RUS}^k$ into aggregate effects, we will use the full GE framework in Section 6.¹⁸ For ease of notation, we drop the RUS subscript for the remainder of this section.

Guided by equation (12), we set up an empirical counterpart that respects this structure and accounts for the temporal dimension in the data. Our empirical design compares (i) fully sanctioned versus never-sanctioned products, (ii) EU (and mirroring) versus non-coalition origins, and (iii) pre- versus post-ban periods. Identification therefore comes from the relative decline in EU exports of fully sanctioned products to Russia after the bans, net of changes for non-sanctioning countries and for products never subject to restrictions. Specifically, the regression specification is given by

$$X_{et}^k = \exp \left[\sum_{m=\text{Feb 2019}}^{\text{Dec 2024}} \beta_m S^k \cdot S_e \cdot \mathbf{1}\{\text{event}_t = m\} + \mu_e^k + \mu_{et} + \mu_t^k \right] + \epsilon_{et}^k, \quad (13)$$

where X_{et}^k denotes Russian imports from origin e of HS6 product k in month t , $S^k = 1$ for fully sanctioned products, and $S_e = 1$ for exporters that are part of the sanctioning coalition. The coefficient path $\{\beta_m\}$ traces the evolution of sanctioned EU export flows relative to the three comparison groups described above. As is standard in the trade literature, we follow Santos Silva and Tenreyro (2006) and estimate equation (13) in its multiplicative form using Poisson Pseudo-Maximum Likelihood (PPML). This avoids biased and inconsistent estimates in the presence of heteroskedasticity due to the log-linearization necessary for OLS estimation and can accommodate zero flows. This feature is

¹⁸ Note that we measure the average change in $\theta^k \log(\tau_{e,RUS}^k)$, which we will translate into $\tau_{e,RUS}^k$ using the trade elasticities, θ^k , described in Section 4.3.

particularly important in our context, as export bans precisely aim to lower the respective flows all the way to zero, and observations corresponding to fully successful bans would be dropped if a log-linear counterpart of (13) were estimated instead.

The specification includes three sets of high-dimensional fixed effects with a clear structural interpretation relating to a multi-period stochastic counterpart of the theoretical gravity equation (12). Product-origin fixed effects, μ_e^k , absorb time-invariant technological differences, T_e^k , determining comparative advantage across and absolute advantage within products, as well as time-invariant geographical and non-sanction-related trade policy determinants of exporter-product-specific trade costs with Russia. Origin-month fixed effects μ_{et} capture the exporter’s competitiveness in terms of its wage, w_{et} , as well as origin-wide trade cost shocks with Russia that affect all products, such as reputational risk or logistical and financial constraints (including SWIFT exclusion).¹⁹ Product-month fixed effects, μ_t^k , absorb time-varying Russian demand at the product level, $X_{RUS,t}^k$, as well as the product-specific competitive environment in the Russian market, as captured by the price index $P_{RUS,t}^k$, or equivalently by the so-called inward multilateral resistance term in a gravity framework, which formalizes how accessible the Russian market for the respective product is overall for suppliers from all origins.

Figure 8 reports the estimated event-study coefficients, normalizing January 2019 to zero. We find no evidence of differential pre-trends. Exports begin to decline when the war starts, but the sharp and persistent drop occurs in July 2022, the first month in which the initial wave of export bans is fully in force. This pattern is consistent with the long transition periods built into the sanctions regime: although the bans were announced earlier, many only became legally binding in mid-2022.²⁰

A useful benchmark is the case of perfect enforcement. Under a gravity structure like ours, a fully enforced export ban corresponds to an infinite bilateral trade cost, which would imply that sanctioned EU exports fall to zero in Russian customs data. In that case, the trade-cost increase would be unidentified. The large—but far from complete—decline observed in the data therefore pins down the finite de facto increase in trade costs implied by the residual flows. Our triple-difference design exploits precisely this remaining variation.

Our baseline estimate implies that the export bans reduced EU exports of fully sanctioned products to Russia by roughly 80 percent, relative to the control flows. This is a substantial decline, but it falls short of the complete cessation implied by perfect enforcement,

¹⁹ Technically, to control for the impact of w_{et} on trade, we would need origin-product-month fixed effects, because w_{et} enters the gravity equation with a product-specific elasticity, θ^k . However, this would absorb all the variation in X_{et}^k .

²⁰ Figure A5 in the Appendix shows the results for the same specification (13) and Figure 8, but for export quantities rather than export values as the dependent variable. The pattern is very similar, with a somewhat stronger average effect size of -85% .

consistent with the extensive evasion documented above. Through the lens of our gravity model, and using the product-specific trade elasticities, this implies that effective bilateral trade costs on EU exports of banned products to Russia increased by about 27% on average. In the following section, we quantify the aggregate effects associated with these DiD results and contrast them with alternative counterfactual scenarios.

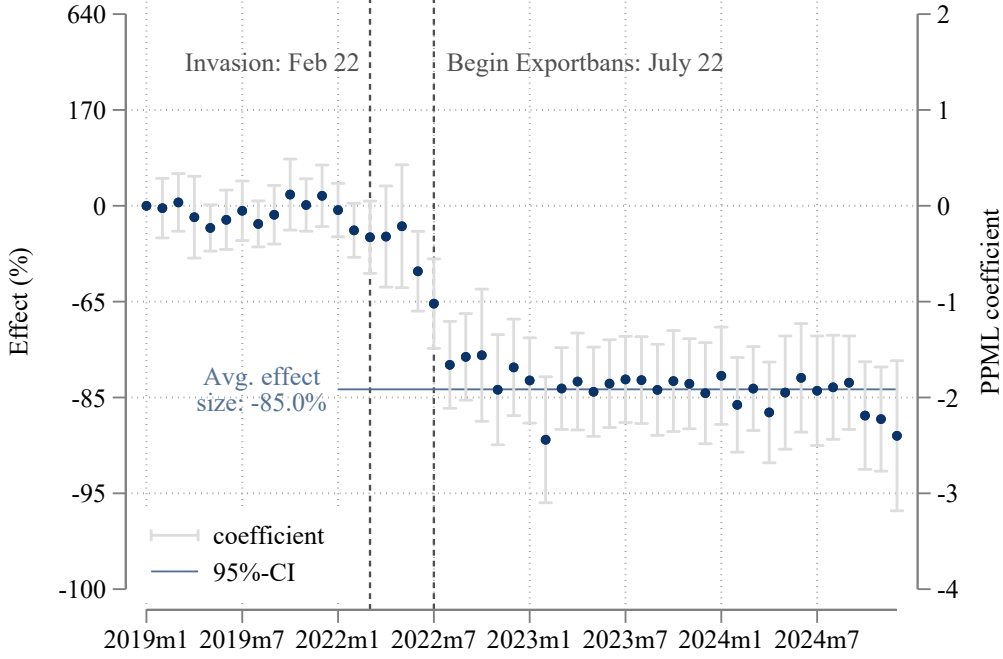


FIGURE 8. BASELINE ESTIMATION RESULTS: EVENT-STUDY

Note: The figure shows the estimated $\hat{\beta}_m$ coefficients of the event-study specification (13). The coefficient for the first month (January 2019) is normalized to zero. On the right y-axis, the raw coefficients are shown. On the left y-axis, they are translated into partial equilibrium trade reductions as $(e^{\hat{\beta}_m} - 1) \times 100\%$.

6 Counterfactual Exercises and Mechanisms

After showing that the export bans reduced Russian imports substantially but not fully, we put our quantitative model from Section 4 to work and compute several counterfactual scenarios. This allows us understand (i) the effective aggregate consequences of the export bans, (ii) how far we are from full enforcement, and (iii) the mechanisms behind the evasion patterns.

6.1 Quantifying the Effects of Sanctions and Evasion

We shock the trade costs of all the products that were fully banned by the end of 2024, taking into account which part of the coalition sanctioned which products (see Section 4.3

TABLE 3. COUNTERFACTUAL SCENARIOS: RUSSIA’S IMPORTS OF BANNED PRODUCTS

Scenario	Change in Imports	% Full Enforcement	% Observed Enforcement
Full Enforcement	-100%		
Observed Evasion Costs	-71%	71%	
Minimum Evasion Costs	-36%	36%	51%
Minimum Evasion Costs (excl. Top 5 Hubs)	-45%	45%	64%

Notes. This table reports how Russia’s imports of banned products from coalition countries change across counterfactual scenarios. In each scenario, we shock the coalition’s export costs of banned products to Russia while varying the transshipment opportunities. Under “Full Enforcement,” transshipment is blocked in the model, so there are no exports of banned products. In the “Observed Evasion Costs” scenario, we increase the export costs of banned products by 27% on average (while blocking transshipment), motivated by our DiD estimate in Section 5. “Minimum Evasion Costs” refers to the case where firms from coalition origins can use non-coalition countries as intermediary hubs to transship to Russia. In “Minimum Evasion Costs (excl. Top 5 Hubs),” we exclude the five hubs that are most used in the minimum-cost scenario. World GDP is the numéraire. The results are discussed in Section 6.

TABLE 4. COUNTERFACTUAL SCENARIOS: RUSSIA’S REAL INCOME

Scenario	$\hat{W}$	% of Full Enforcement	% of Observed Enforcement
Full Enforcement	-1.20%		
Observed Evasion Costs	-0.93%	77%	
Minimum Evasion Costs	-0.39%	32%	42%
Minimum Evasion Costs (excl. Top 5 Hubs)	-0.49%	41%	53%
Maximum Harm	-2.98%	247%	320%

Notes. This table reports Russia’s real income changes, denoted by $\hat{W}$, across counterfactual scenarios. In the “Maximum Harm” scenario, coalition countries stop exporting to Russia entirely (and transshipment opportunities are blocked). For details on the remaining scenarios, see the notes to Table 3. The results are discussed in Section 6.

for details). In certain scenarios, we allow for transshipment of the coalition’s exports. Coalition countries cannot be used as transshipment hubs, even for products that they did not ban.²¹ Finally, the export bans that the EU imposed on Belarus are fully enforced throughout, such that it can never be used as a hub for these products (see Section 4.3 for details). To be consistent with the theory, we first solve the equilibrium to have balanced trade in the baseline levels.

We quantify four counterfactual scenarios. The results are shown in Tables 3 and 4 for Russia. Table 5 reports the corresponding EU aggregate impact on exporters. First, as a benchmark, export bans are fully enforced and cannot be rerouted. This is the implicit assumption in the standard quantitative literature that evaluates the (potential) consequences of export bans and large trade-cost increases more generally. This yields the maximum changes (and losses) achievable under the current set of sanctions in the model. Mechanically, exports from coalition countries drop to zero because the export costs are prohibitive and cannot be circumvented. This leads to a real income—or real

²¹ We treat Ukraine as part of the coalition but assume its export bans are fully enforced and that it never acts as a transshipment hub.

GDP—loss of 1.20% for Russia. While it is well-known that these types of models typically yield relatively small changes in real income (see [Costinot and Rodríguez-Clare, 2014](#), for an overview),²² evaluating the relative magnitudes across scenarios is instructive to understand which channels matter most through the lens of the model. Moreover, even though real income changes correspond to welfare changes in our model, sanctions are typically motivated by other reasons than just income effects, which is why we will explicitly call them real income losses throughout.

In the second counterfactual scenario, we translate the observed effect from the DiD in Section 5 into a trade-cost shock. Specifically, we increase the trade costs of banned products by 27% on average, which comes from $\exp(-1.684)^{(-1/\theta^k)}$, where -1.684 is the static DiD estimate and θ^k is the product-specific trade elasticity. Further transshipment is blocked in this scenario to ensure that effective trade costs increase by the direct shock size. The second row of Table 3 shows that Russian imports of banned products drop by 71%. Note that this is close to the magnitude we see in the raw data toward the end of 2024 (see Figure 4). Hence, imports drop by 29 pp less than under full enforcement. Table 4 shows that this translates into a real income loss of 0.93%. Accordingly, while evasion is costly in reality, it reduced the full-enforcement impact of the export bans—that we obtain in standard quantitative analyses—by 23%. Put differently, from a policy perspective, fully enforcing the export bans would have increased Russia’s income losses by about 30%.

In the third counterfactual scenario, we study evasion: export bans can be circumvented via transshipment. Recall that the only costs of evasion in the model are the additional trade costs from involving a transshipment hub. Russia’s imports of banned products drop by 39% in this case (Table 3, third row). That is, even when allowing for full transshipment, imports of banned products drop substantially because re-exporting entails additional costs. Moreover, the large coalition size implies that many potential hubs cannot be used for transshipment. However, a substantial gap remains between this scenario and the observed enforcement level. This is likely because in reality, there is an additional concealment cost and a risk of detection that add to the transshipment burden—more on the potential mechanism behind this gap below. Note that trade costs are held constant for non-banned products, even though trading with Russia became more difficult across the board, and an increase in all the trade costs to Russia would also predict larger transshipment costs for banned products.

Russia’s real income losses under ‘full’ evasion amount to 0.40% of the baseline level, substantially smaller than under the observed evasion costs. That is, in reality, implicit or explicit efforts to curb evasion increased the losses by about 130% (compare the second

²² As an example, using a similar framework, [Caliendo and Parro \(2015\)](#) find that U.S. welfare increased by 0.08% following NAFTA’s tariff reductions.

TABLE 5. COUNTERFACTUAL SCENARIOS: EU EXPORTS OF BANNED PRODUCTS TO RUSSIA AND REAL INCOME

Country	Baseline Exp (USD in M)	Banned Exports to Russia		Change in Real Income, $\hat{W}$	
		Observed Transshipment	No Transshipment	Observed Transshipment	No Transshipment
EU	58,838	-75%	-100%	-0.025%	-0.035%

Notes. This table reports EU aggregate changes in total exports of banned products to Russia and the implied real-income changes across the three main counterfactual scenarios. Export changes are relative to the baseline flows. “Full Enforcement,” “Observed Evasion Costs,” and “Minimum Evasion Costs” correspond to the scenarios in Tables 3 and 4. EU real income changes are the simple average of member-state real-income changes.

and third rows in Table 4). Moreover, the model suggests that closing a handful of hubs has a significant impact: if we block transshipment through the five main hubs in the model, imports drop by an additional 8 pp, while the losses increase by about 25%. In Section 6.2, we confront the main hubs in the model and the data.

Finally, we compute a scenario in which the coalition countries stop exporting entirely to Russia, while blocking transshipment opportunities. This tells us, through the lens of the model, the maximum achievable harm. Table 4 shows that this scenario more than triples Russia’s real income losses relative to the current regime.

Impact on EU exporters. So far, we have focused on the consequences for Russia. Table 5 reports the corresponding effects on the EU, in terms of (i) exports of banned products to Russia and (ii) real income. Under “Full Enforcement,” EU exports of banned products to Russia collapse by construction, whereas under “Observed Evasion Costs” and “Minimum Evasion Costs” they are partially preserved. Total EU exports of banned products drop by 68% under the observed (DiD) trade-cost shock, while they fall by 41% in the full transshipment scenario. The real-income effects are modest relative to the losses borne by Russia across all counterfactual exercises, with EU members losing on average 0.030% of real income under the current sanction regime. However, Appendix Table A2 highlights substantial heterogeneity across member states, with, for instance, Lithuania’s real income falling by about 0.17%. Note that, mirroring the results for Russia, the real income losses would be larger under full enforcement.

6.2 Potential Mechanisms behind Transshipment

In this section, we investigate the model’s performance in explaining the data as well as the potential determinants of which transshipment hubs are selected. First, we confront the predictions of the model with the hubs in the customs data. Then, we study what could potentially explain the gaps between the empirical and the model-implied transshipment

patterns. We restrict the sample to origins from the EU plus those that mirror the EU’s sanctions (cf. Table 1), because for the remaining coalition countries, our sanctions data do not cleanly identify fully vs. partially sanctioned products, as explained in Section 4.3.

Model predictions vs. data. Figure 9 compares the transshipment hubs for banned products observed in the customs data with the model’s predictions. In the model, only the minimum-cost hub is selected by an origin-product cell. In the customs data, certain origin-product cells use multiple hubs, so we divide the number of origin-products using a specific hub by the total number of origin-product-hub cells with positive flows.

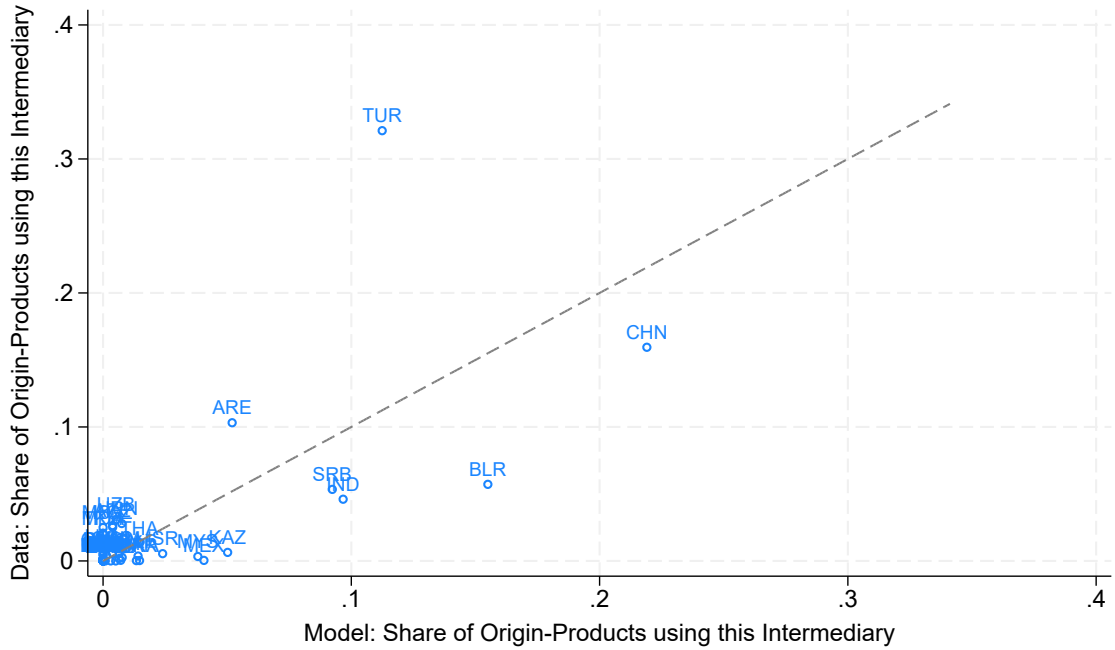


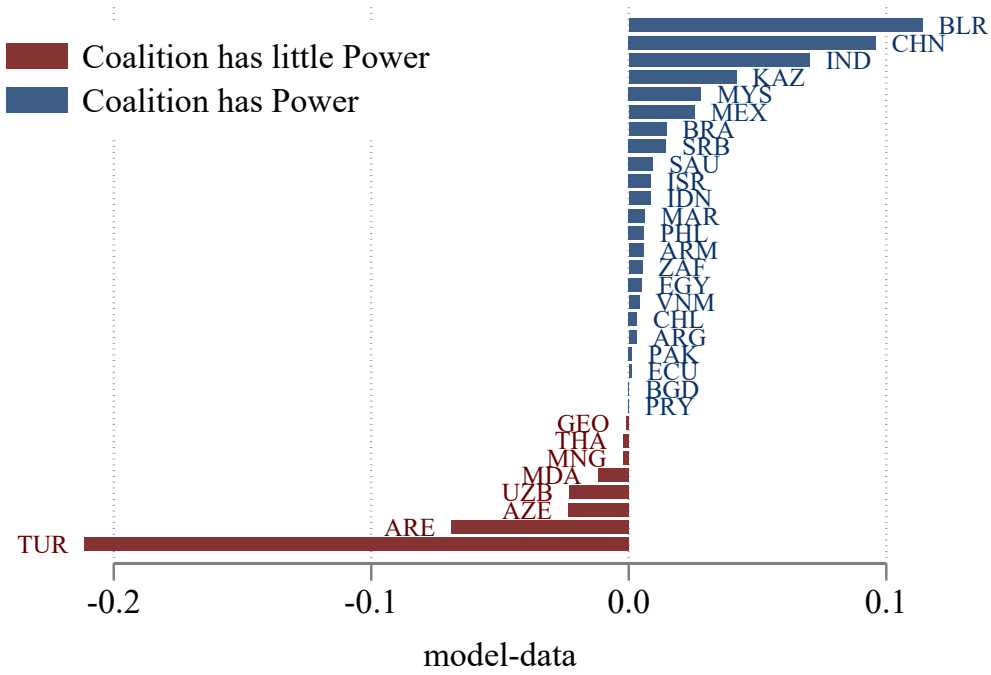
FIGURE 9. TRANSSHIPMENT HUBS: MODEL PREDICTION VS. DATA

Note: This figure compares the transshipment hubs in the Russian customs data with the model’s prediction on which hubs are selected after the direct export bans. The sample uses only products hit by an export ban. For the data share, we use the year 2024 and divide the number of origin-product cells that use a specific hub by the total number of origin-product-hub cells with positive flows. The model share uses, for each origin-product cell, the minimum-cost hub (as only a single hub is selected by an origin for a given product in the model). The sample is restricted to EU (plus mirroring) origins and non-coalition dispatch countries. The dashed line is a 45°-line. Section 6.2 discusses the results.

Several interesting insights emerge. Despite the simplicity of our modeling of transshipment, the framework’s predicted hubs are consistent with the data. Importantly, this is a demanding out-of-sample test, as the model uses only trade data from 2019 and does not rely on Russian customs data. Both the data and the model predict that transshipment is concentrated among a handful of hubs. The five main hubs in the data are Belarus, China, Serbia, Türkiye, and the United Arab Emirates, while in the model, India replaces the UAE. The combined shares of the main five hubs are 69% and 68%, respectively, in

the data and in the model. The correlation between hub use in the model and the data is 0.70 (and 0.83 if we exclude Türkiye).

Overall, this reveals that using fundamental trade costs to predict transshipment hubs is powerful. However, China and Belarus are substantially more attractive hubs in the model, while Türkiye is by far the most used hub in the data. We next look at what could potentially explain the gaps between the data and the model.



(A) GEOPOLITICAL POWER OF THE COALITION?



(B) LOGISTICS?

FIGURE 10. THE POTENTIAL DRIVERS OF DATA-MODEL GAPS IN HUB SELECTION

Note: This figure investigates the potential drivers of the difference between how often a hub is selected in the model and in the data (i.e., the difference between the horizontal and the vertical axes in Figure 9). Panel 10a sorts the hubs by their model-data gaps and categorizes them as countries where the coalition has geopolitical power if the gap is positive (i.e., when the model overpredicts hub use), and with little power otherwise. Panel 10b plots how often a hub is used empirically for banned vs. non-sanctioned products in 2024. The dashed line is a 45°-line. Section 6.2 discusses the results.

Drivers of model-data gaps. The differences between hub use in the data and the model can be interpreted as additional forces that determine transshipment opportunities and that are currently not picked up in our quantification. Figure 10 investigates two potential drivers.

It is natural to think that geopolitical alignment and, hence, an implicit bargaining power of the coalition over potential hubs could be important in this context. However, as Panel 10a highlights, this is unlikely to explain the model-data gaps, because we would need to classify countries such as Belarus, China, and Kazakhstan as being among the hubs where coalition countries have higher bargaining power than over countries like Serbia, Uzbekistan, and Türkiye.

By contrast, Panel 10b reveals that there seems to be a major underlying force driving the choice of a transshipment hub that appears to be related to general logistics. That is, the share of origin-product cells that use a hub empirically for fully sanctioned (banned) products is almost identical to the share that uses that hub to ship non-sanctioned products (correlation of 0.99). This implies that knowing how coalition countries legally export goods to Russia is highly predictive of how illegal flows take place. This, in turn, may suggest the presence of economies of scope from shipping multiple different products via a hub. Such spillovers are not present in our framework, as is standard in the literature, but may help explain the gap between our model-implied shares and those in the data.

7 Conclusion

We ask how exporters evade export bans and how evasion shapes the effectiveness of sanctions. We focus on the EU’s export bans on Russia after the 2022 invasion of Ukraine, combining a newly collected product-level dataset on the legal scope and timing of the bans with Russian transaction-level customs data that record both the true origin and the shipping (dispatch) country of each transaction. This setting allows us to directly observe evasion through third-country transshipment. We show that at least 26% of pre-war exports of fully banned EU products continue to reach Russia, transshipped via a small set of intermediary hubs.

To quantify the consequences of these flows and understand the drivers of evasion, we build a multi-product EK model with endogenous transshipment. Calibrated at the most disaggregated product level available, the model separates legal diversion to non-sanctioning suppliers from evasion via hubs. We then use a triple DiD (gravity) setup to recover the de facto increase in trade costs implied by the bans. The estimates indicate that the export bans raised effective trade costs from coalition countries to Russia by around 27% on average, far below the prohibitive levels assumed under full enforcement.

In the model, this observed level of evasion reduces Russia’s real income loss by 23% compared with a full-enforcement scenario. Put differently, from a policy perspective, fully enforcing the export bans would have increased Russia’s income losses by about 30%. A “minimum-cost” scenario in which evasion is limited only by additional transport costs yields even smaller losses, implying that efforts to curb re-exports restricted transshipment opportunities.

Finally, we show that fundamental trade costs go a long way in predicting which countries become transshipment hubs, but cannot fully account for the data-model gaps. For instance, Türkiye is used much more often in the data than the model predicts. We argue that the main driver of the data-model gaps is unlikely to be geopolitical alignment but rather seems related to general logistics, as there is a near one-to-one mapping between a hub’s role in legal and illegal trade.

Taken together, our results suggest that the design and enforcement of sanctions (and trade policy in general) cannot necessarily be evaluated under the assumption of perfect enforcement. While our focus is on export bans, the quantitative framework we develop can be applied more broadly to settings in which trade barriers might be circumvented through third countries, including tariff wars and other instances of targeted protectionism.

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A Data

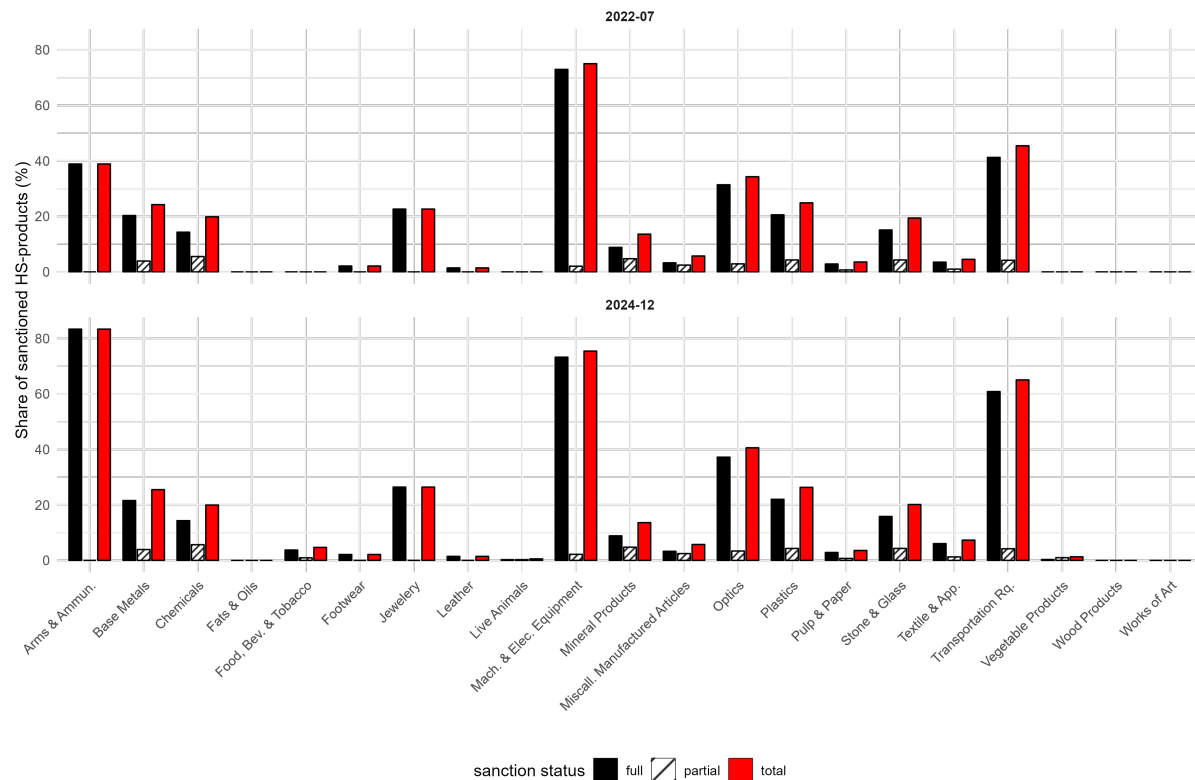


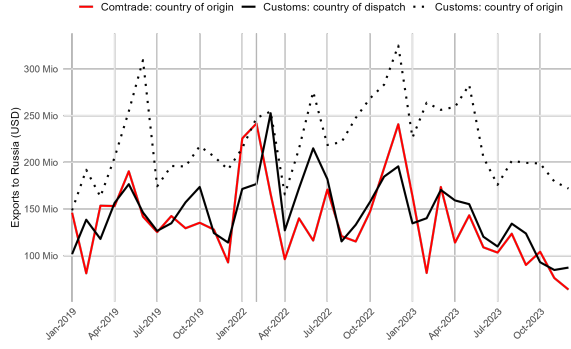
FIGURE A1. SECTORAL COVERAGE OF EU-BLR SANCTIONS BY STATUS AND EIF DATE

Note: This figure reports, by sector, the share of EU-sanctioned products vis-a-vis Belarus at the 6-digit HS2017 level, distinguished by sanction status. Black bars indicate the share of fully sanctioned products, striped bars the share of partially sanctioned products, and red bars the total share of sanctioned products. The upper panel includes all sanctions that came into force by July 2022, while the lower panel includes all sanctions in effect by December 2024.

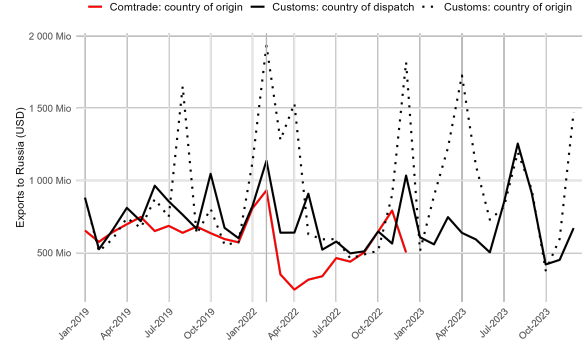
1st package	Feb, 23 th 2022	-
2nd package	Feb, 25 th 2022	dual-use/high-tech*, oil technologies*, aviation*
3rd package	March, 1 st 2022	banknotes*
compliance	March, 9 th 2022	maritime navigation*
4th package	March, 15 th 2022	steel and iron* †, luxury*
5th package	April, 8 th 2022	goods strength. the ind. capability*, aviation*, luxury*, oil technologies*, dual-use/high-tech*, coal†, strategic†, steel & iron†
6th package	June, 3 rd 2022	dual-use/high-tech*, petroleum oils†, strategic†
maintenance & alignment	July, 21 st 2022	dual-use/high-tech*, goods strengthening the ind. capability*, oil technologies*, gold†
8th package	Oct, 10 th 2022	aviation*, dual-use/high-tech*, firearms*, goods strengthening the ind. capability*, iron and steel†, strategic†
9th package	Dec, 16 th 2022	dual-use/high-tech*, aviation*, goods strengthening the ind. capability*, petroleum oils†, iron and steel†
10th package	Feb, 25 th 2023	dual-use/high-tech*, aviation*, goods strengthening the ind. capability*, strategic†
11th package	June, 23 rd 2023	dual-use/high-tech*, firearms*, goods strengthening the ind. capability*, luxury*, strategic†, steel and iron†
12th package	Dec, 18 th 2023	dual-use/high-tech*, goods strengthening the ind. capability*, strategic†, diamonds†
13th package	Feb, 23 rd 2024	dual-use/high-tech*, goods strengthening the ind. capability*
14th package	June, 24 th 2024	dual-use/high-tech*, goods strengthening the ind. capability*, strategic†

TABLE A1. TIMELINE OF EU SANCTIONS PACKAGES AGAINST RUSSIA.

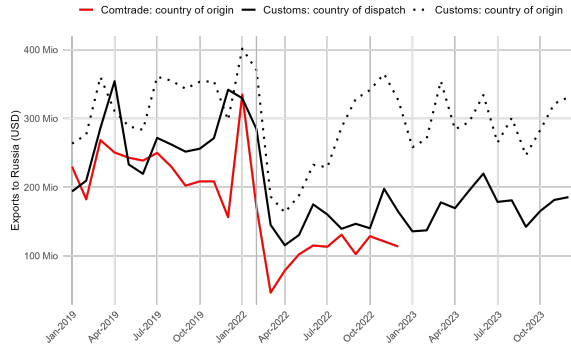
Note: * export sanction, † import sanction



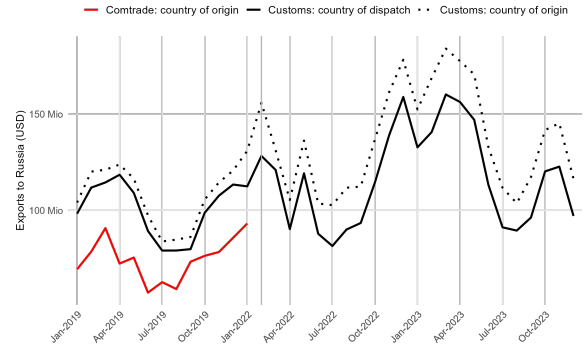
(A) BRAZIL



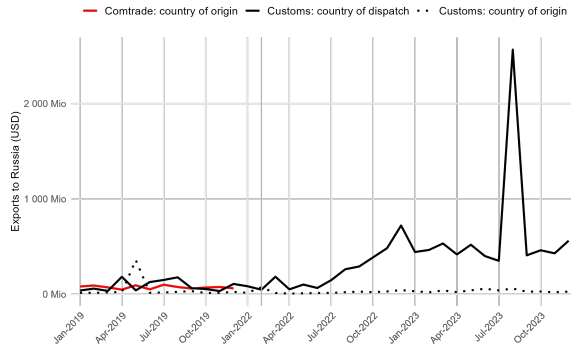
(B) SOUTH KOREA



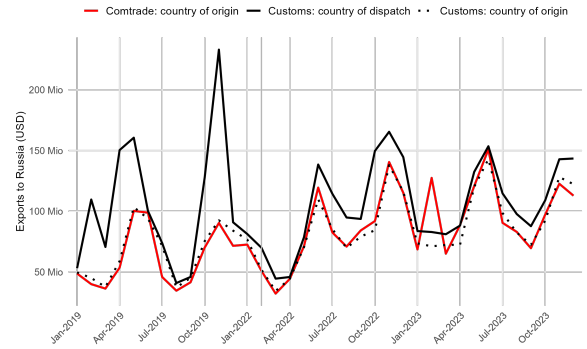
(C) VIETNAM



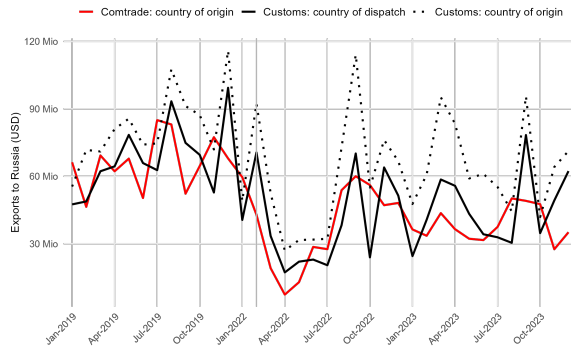
(D) ECUADOR



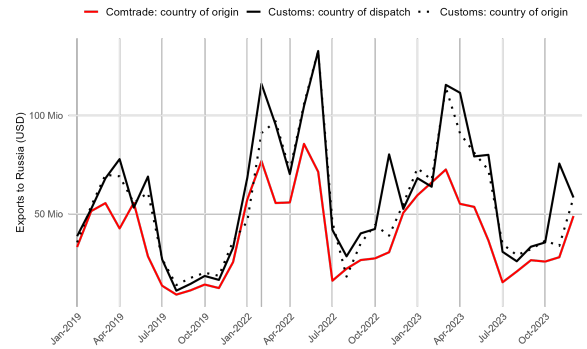
(E) UNITED ARAB EMIRATES



(F) AZERBAIJAN



(G) CHILE



(H) EGYPT

Note: The figure shows Russian imports from the customs data (by month, year, and country of dispatch & origin) alongside exports to Russia from the same countries reported in UN Comtrade. Data are shown for 2019 (pre-period) and 2022–2023 (post-onset of the Russian war) for other non-EU, non-coalition exporters with available Comtrade data. Trade values are in millions of US dollars.

FIGURE A2. VALIDATION OF RUSSIAN CUSTOMS DATA (OTHER EXAMPLES)

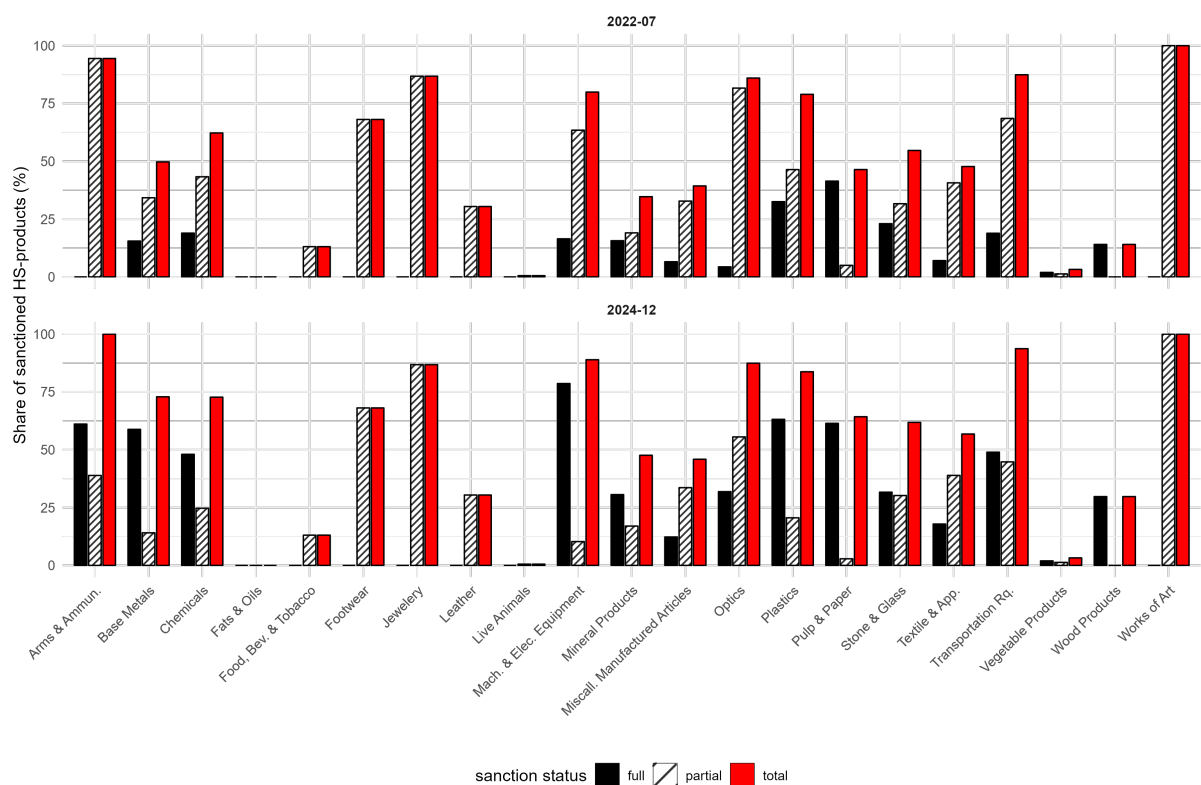


FIGURE A3. EU EXPORT BANS BY SECTORS

Note: The figure reports, by sector, the share of HS6 products covered by EU export bans. Black bars indicate fully banned products, striped bars partially banned products, and red bars total banned products. The upper panel shows coverage as of July 2022, and the lower panel as of December 2024.

B Additional Results

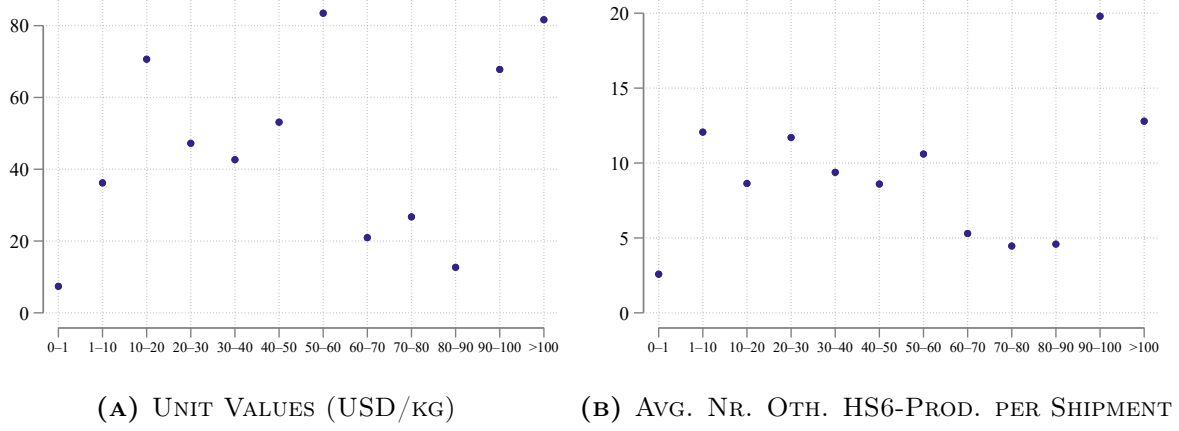


FIGURE A4. EVASION AND PRE-WAR CHARACTERISTICS OF EXPORT-BANNED PRODUCTS

Note: Bins are constructed over an evasion measure defined for each HS6 product as the ratio of its average monthly export value from the EU to Russia in the post-export-ban period (August 2022–December 2024) to its average monthly export value from the EU to Russia in 2019. The sample consists of 668 HS6 products that became subject to EU export bans after July 2022. Each point reports the unweighted mean of the characteristic shown on the y-axis within a bin of the evasion measure, except for unit values, which are reported as medians within bins. All product characteristics refer to 2019.

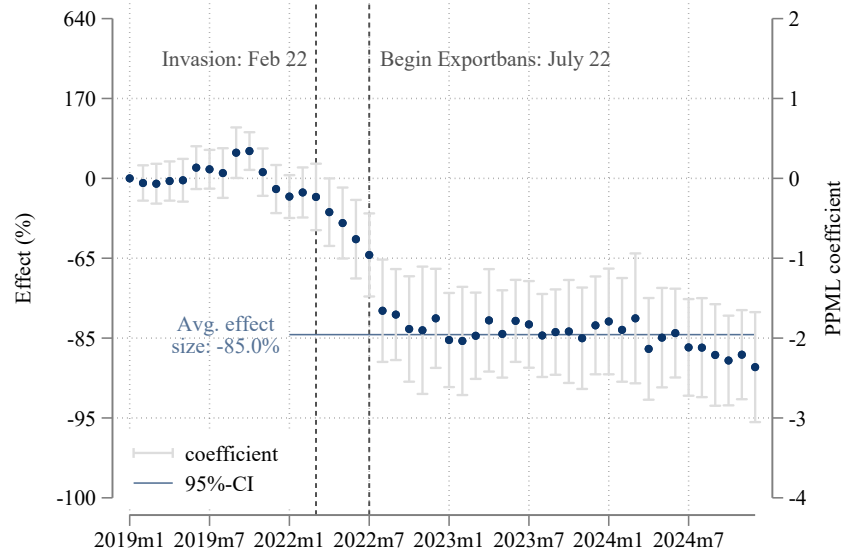


FIGURE A5. EVENT-STUDY ESTIMATION RESULTS (QUANTITIES)

Note: The figure shows the estimated $\hat{\beta}_m$ coefficients of the event-study specification (13), but using export quantities rather than values as the dependent variable. The coefficient for the first month (January 2019) is normalized to zero. On the right y-axis, the raw coefficients are shown. On the left y-axis, they are translated into partial equilibrium trade costs reductions as $(e^{\hat{\beta}^m} - 1) \times 100\%$.

TABLE A2. COUNTERFACTUAL SCENARIOS: EU MEMBER-STATE EXPORTS OF BANNED PRODUCTS TO RUSSIA AND REAL INCOME

Country	Baseline Exp (USD in M)	Banned Exports to Russia		Change in Real Income, $\hat{W}$	
		Observed Transshipment	No Transshipment	Observed Transshipment	No Transshipment
EU	58,838	-75%	-100%	-0.025%	-0.035%
AUT	1,656	-79%	-100%	-0.018%	-0.021%
BEL	2,140	-71%	-100%	-0.011%	-0.016%
BGR	298	-73%	-100%	-0.028%	-0.038%
CYP	39	-88%	-100%	-0.015%	-0.014%
CZE	2,930	-71%	-100%	-0.065%	-0.094%
DEU	16,385	-76%	-100%	-0.031%	-0.041%
DNK	583	-77%	-100%	-0.015%	-0.019%
ESP	1,718	-76%	-100%	-0.008%	-0.012%
EST	1,149	-77%	-100%	-0.238%	-0.304%
FIN	3,470	-79%	-100%	-0.128%	-0.157%
FRA	5,768	-71%	-100%	-0.016%	-0.023%
GRC	227	-47%	-100%	-0.014%	-0.023%
HRV	78	-76%	-100%	-0.008%	-0.008%
HUN	1,167	-79%	-100%	-0.034%	-0.045%
IRL	142	-77%	-100%	0.004%	0.005%
ITA	5,226	-78%	-100%	-0.025%	-0.031%
LTU	2,420	-83%	-100%	-0.333%	-0.394%
LUX	237	-78%	-100%	-0.030%	-0.036%
LVA	634	-56%	-100%	-0.097%	-0.163%
MLT	25	-50%	-100%	-0.101%	-0.126%
NLD	4,294	-67%	-100%	-0.050%	-0.070%
POL	4,265	-77%	-100%	-0.056%	-0.077%
PRT	245	-85%	-100%	-0.005%	-0.005%
ROU	783	-67%	-100%	-0.024%	-0.035%
SVK	679	-74%	-100%	0.017%	0.019%
SVN	549	-83%	-100%	-0.057%	-0.068%
SWE	1,731	-75%	-100%	-0.020%	-0.026%

Notes. Same as Table 5 but extended to include the single member countries.